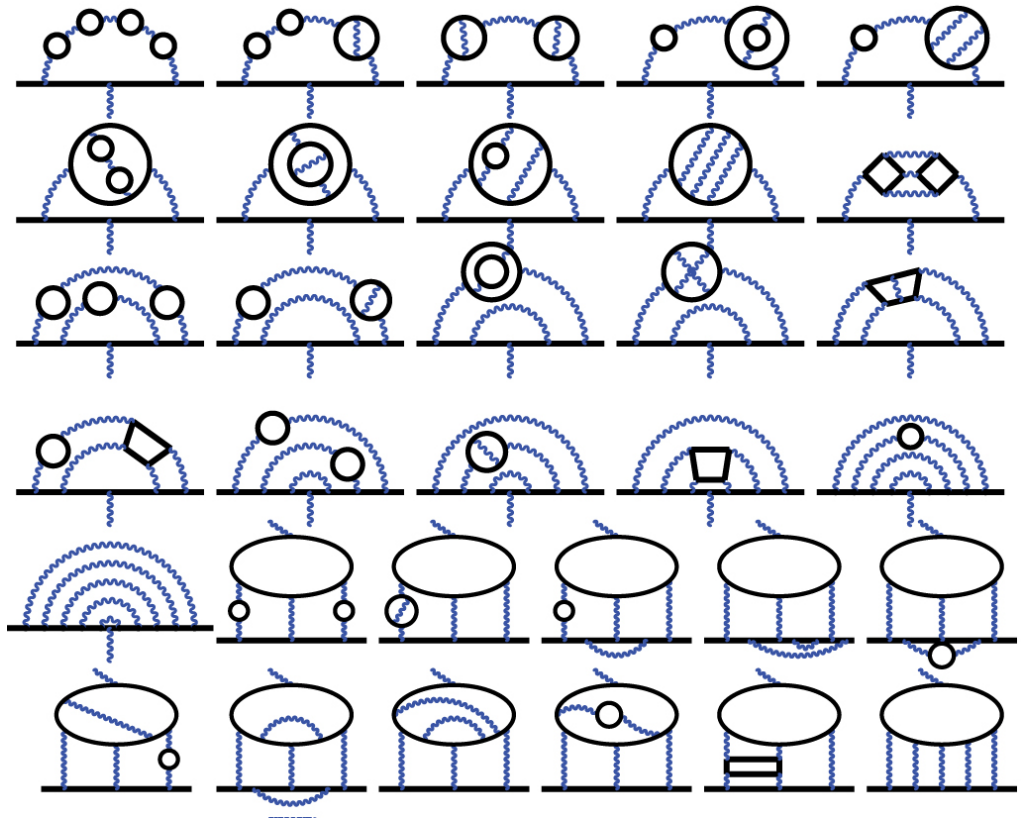
Overview of Generators and Pythia8

Stephen Mrenna
Fermilab

October 17, 2023

Many sciences: highly mathematized
Particle physics: QFT: extremely successful



(g-2) electron
predicted = 0.001 159 652 181 643 (764)
observed = 0.001 159 652 180 59 (13)

Good enough to be on a coffee mug



†

†Yvonne Tang, SLAC National Accelerator Laboratory

Practitioner's "SM"

$$-\frac{1}{2}\partial_\nu g^a_\mu \partial_\nu g^a_\mu - g_s f^{abc}\partial_\mu g^a_\nu g^b_\mu g^c_\nu - \frac{1}{4} g_s^2 f^{abc} f^{ade} g^b_\mu g^c_\nu g^d_\mu g^e_\nu + \frac{1}{2} i g_s^2 (\bar{q}^\sigma_i \gamma^\mu q^\sigma_j) g^a_\mu + \bar{G}^a \partial^2 G^a + g_s f^{abc} \partial_\mu \bar{G}^a G^b g^c_\mu - \partial_\nu W^+_\mu \partial_\nu W^-_\mu -$$

$$M^2 W^+_\mu W^-_\mu - \frac{1}{2}\partial_\nu Z^0_\mu \partial_\nu Z^0_\mu - \frac{1}{2c_w^2} M^2 Z^0_\mu Z^0_\mu - \frac{1}{2}\partial_\mu A_\nu \partial_\mu A_\nu - \frac{1}{2}\partial_\mu H \partial_\mu H - \frac{1}{2} m_h^2 H^2 - \partial_\mu \phi^+ \partial_\mu \phi^- - M^2 \phi^+ \phi^- - \frac{1}{2}\partial_\mu \phi^0 \partial_\mu \phi^0 -$$

$$\frac{1}{2c_w^2} M \phi^0 \phi^0 - \beta_h [\frac{2M^2}{g^2} + \frac{2M}{g} H + \frac{1}{2}(H^2 + \phi^0 \phi^0 + 2\phi^+ \phi^-)] + \frac{2M^4}{g^2}\alpha_h - igc_w [\partial_\nu Z^0_\mu (W^+_\mu W^-_\nu - W^+_\nu W^-_\mu) - Z^0_\nu (W^+_\mu \partial_\nu W^-_\mu - W^-_\mu \partial_\nu W^+_\mu) +$$

$$Z^0_\mu (W^+_\nu \partial_\nu W^-_\mu - W^-_\nu \partial_\nu W^+_\mu)] - igs_w [\partial_\nu A_\mu (W^+_\mu W^-_\nu - W^+_\nu W^-_\mu) - A_\nu (W^+_\mu \partial_\nu W^-_\mu - W^-_\mu \partial_\nu W^+_\mu) + A_\mu (W^+_\nu \partial_\nu W^-_\mu - W^-_\nu \partial_\nu W^+_\mu)] -$$

$$\frac{1}{2} g^2 W^+_\mu W^-_\mu W^+_\nu W^-_\nu + \frac{1}{2} g^2 W^+_\mu W^-_\nu W^+_\mu W^-_\nu + g^2 c_w^2 (Z^0_\mu W^+_\mu Z^0_\nu W^-_\nu - Z^0_\mu Z^0_\mu W^+_\nu W^-_\nu) + g^2 s_w^2 (A_\mu W^+_\mu A_\nu W^-_\nu - A_\mu A_\mu W^+_\nu W^-_\nu) +$$

$$g^2 s_w c_w [A_\mu Z^0_\nu (W^+_\mu W^-_\nu - W^+_\nu W^-_\mu) - 2A_\mu Z^0_\mu W^+_\nu W^-_\nu] - g\alpha [H^3 + H\phi^0\phi^0 + 2H\phi^+\phi^-] - \frac{1}{8} g^2 \alpha_h [H^4 + (\phi^0)^4 + 4(\phi^+\phi^-)^2 + 4(\phi^0)^2 \phi^+\phi^- +$$

$$4H^2\phi^+\phi^- + 2(\phi^0)^2 H^2] - gMW^+_\mu W^-_\mu H - \frac{1}{2} g \frac{M}{c_w^2} Z^0_\mu Z^0_\mu H - \frac{1}{2} ig [W^+_\mu (\phi^0 \partial_\mu \phi^- - \phi^- \partial_\mu \phi^0) - W^-_\mu (\phi^0 \partial_\mu \phi^+ - \phi^+ \partial_\mu \phi^0)] + \frac{1}{2} g [W^+_\mu (H\partial_\mu \phi^- -$$

$$\phi^- \partial_\mu H) - W^-_\mu (H\partial_\mu \phi^+ - \phi^+ \partial_\mu H)] + \frac{1}{2} g \frac{1}{c_w} (Z^0_\mu (H\partial_\mu \phi^0 - \phi^0 \partial_\mu H) - ig \frac{s_w^2}{c_w} M Z^0_\mu (W^+_\mu \phi^- - W^-_\mu \phi^+) + igs_w M A_\mu (W^+_\mu \phi^- - W^-_\mu \phi^+) -$$

$$ig \frac{1-2c_w^2}{2c_w} Z^0_\mu (\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) + igs_w A_\mu (\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) - \frac{1}{4} g^2 W^+_\mu W^-_\mu [H^2 + (\phi^0)^2 + 2\phi^+\phi^-] - \frac{1}{4} g^2 \frac{1}{c_w^2} Z^0_\mu Z^0_\mu [H^2 + (\phi^0)^2 +$$

$$2(2s_w^2 - 1)^2 \phi^+\phi^-] - \frac{1}{2} g^2 \frac{s_w^2}{c_w} Z^0_\mu \phi^0 (W^+_\mu \phi^- + W^-_\mu \phi^+) - \frac{1}{2} ig^2 \frac{s_w^2}{c_w} Z^0_\mu H (W^+_\mu \phi^- - W^-_\mu \phi^+) + \frac{1}{2} g^2 s_w A_\mu \phi^0 (W^+_\mu \phi^- + W^-_\mu \phi^+) +$$

$$\frac{1}{2} ig^2 s_w A_\mu H (W^+_\mu \phi^- - W^-_\mu \phi^+) - g^2 \frac{s_w}{c_w} (2c_w^2 - 1) Z^0_\mu A_\mu \phi^+ \phi^- - g^1 s_w^2 A_\mu A_\mu \phi^+ \phi^- - \bar{e}^\lambda (\gamma\partial + m^\lambda_e) e^\lambda - \bar{\nu}^\lambda \gamma\partial \nu^\lambda - \bar{u}^\lambda_j (\gamma\partial + m^\lambda_u) u^\lambda_j - \bar{d}^\lambda_j (\gamma\partial +$$

$$m^\lambda_d) d^\lambda_j + igs_w A_\mu [-(\bar{e}^\lambda \gamma^\mu e^\lambda) + \frac{2}{3}(\bar{u}^\lambda_j \gamma^\mu u^\lambda_j) - \frac{1}{3}(\bar{d}^\lambda_j \gamma^\mu d^\lambda_j)] + \frac{ig}{4c_w} Z^0_\mu [(\bar{\nu}^\lambda \gamma^\mu (1+\gamma^5) \nu^\lambda) + (\bar{e}^\lambda \gamma^\mu (4s_w^2 - 1 - \gamma^5) e^\lambda) + (\bar{u}^\lambda_j \gamma^\mu (\frac{4}{3} s_w^2 - 1 -$$

$$\gamma^5) u^\lambda_j) + (\bar{d}^\lambda_j \gamma^\mu (1 - \frac{8}{3} s_w^2 - \gamma^5) d^\lambda_j)] + \frac{ig}{2\sqrt{2}} W^+_\mu [(\bar{\nu}^\lambda \gamma^\mu (1+\gamma^5) e^\lambda) + (\bar{u}^\lambda_j \gamma^\mu (1+\gamma^5) C_{\lambda\kappa} d^\kappa_j)] + \frac{ig}{2\sqrt{2}} W^-_\mu [(\bar{e}^\lambda \gamma^\mu (1+\gamma^5) \nu^\lambda) + (\bar{d}^\kappa_j C^\dagger_{\lambda\kappa} \gamma^\mu (1 +$$

$$\gamma^5) u^\lambda_j)] + \frac{ig}{2\sqrt{2}} \frac{m^\lambda_e}{M} [-\phi^+ (\bar{\nu}^\lambda (1-\gamma^5) e^\lambda) + \phi^- (\bar{e}^\lambda (1+\gamma^5) \nu^\lambda)] - \frac{g}{2} \frac{m^\lambda_e}{M} [H(\bar{e}^\lambda e^\lambda) + i\phi^0 (\bar{e}^\lambda \gamma^5 e^\lambda)] + \frac{ig}{2M\sqrt{2}} \phi^+ [-m^\kappa_d (\bar{u}^\lambda_j C_{\lambda\kappa} (1-\gamma^5) d^\kappa_j) +$$

$$m^\lambda_u (\bar{u}^\lambda_j C_{\lambda\kappa} (1+\gamma^5) d^\kappa_j] + \frac{ig}{2M\sqrt{2}} \phi^- [m^\lambda_d (\bar{d}^\lambda_j C^\dagger_{\lambda\kappa} (1+\gamma^5) u^\kappa_j) - m^\kappa_u (\bar{d}^\lambda_j C^\dagger_{\lambda\kappa} (1-\gamma^5) u^\kappa_j] - \frac{g}{2} \frac{m^\lambda_u}{M} H (\bar{u}^\lambda_j u^\lambda_j) - \frac{g}{2} \frac{m^\lambda_d}{M} H (\bar{d}^\lambda_j d^\lambda_j) +$$

$$\frac{ig}{2} \frac{m^\lambda_u}{M} \phi^0 (\bar{u}^\lambda_j \gamma^5 u^\lambda_j) - \frac{ig}{2} \frac{m^\lambda_d}{M} \phi^0 (\bar{d}^\lambda_j \gamma^5 d^\lambda_j) + \bar{X}^+ (\partial^2 - M^2) X^+ + \bar{X}^- (\partial^2 - M^2) X^- + \bar{X}^0 (\partial^2 - \frac{M^2}{c_w^2}) X^0 + \bar{Y} \partial^2 Y + igc_w W^+_\mu (\partial_\mu \bar{X}^0 X^- -$$

$$\partial_\mu \bar{X}^+ X^0) + igs_w W^+_\mu (\partial_\mu \bar{Y} X^- - \partial_\mu \bar{X}^+ Y) + igc_w W^-_\mu (\partial_\mu \bar{X}^- X^0 - \partial_\mu \bar{X}^0 X^+) + igs_w W^-_\mu (\partial_\mu \bar{X}^- Y - \partial_\mu \bar{Y} X^+) + igc_w Z^0_\mu (\partial_\mu \bar{X}^+ X^+ -$$

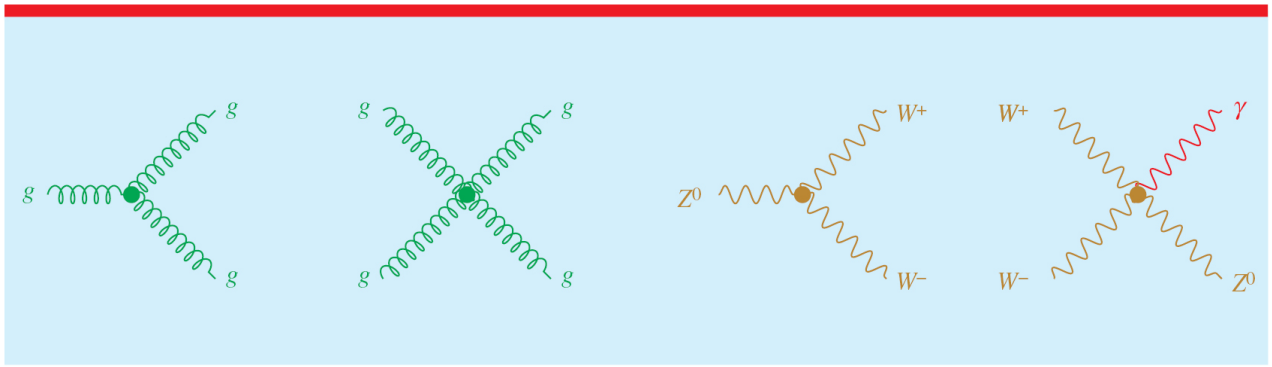
$$\partial_\mu \bar{X}^- X^-) + igs_w A_\mu (\partial_\mu \bar{X}^+ X^+ - \partial_\mu \bar{X}^- X^-) - \frac{1}{2} gM [\bar{X}^+ X^+ H + \bar{X}^- X^- H + \frac{1}{c_w^2} \bar{X}^0 X^0 H] + \frac{1-2c_w^2}{2c_w} igM [\bar{X}^+ X^0 \phi^+ - \bar{X}^- X^0 \phi^-] +$$

$$\frac{1}{2c_w} igM [\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-] + igMs_w [\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-] + \frac{1}{2} igM [\bar{X}^+ X^+ \phi^0 - \bar{X}^- X^- \phi^0]$$

How do we make Predictions?

$\mathcal{L}(x, \partial x) \rightarrow \mathcal{L}(\text{field}, \partial \text{ field})$

$\mathcal{L} \rightarrow$ Feynman rules (QFT)



Feynman rules $\rightarrow$ Feynman graphs (perturbation theory)

small parameter: strong coupling

Smallest power of coupling: leading order (LO)

One more power: next-to-leading order (NLO)

More complex [(tameable) infinities], more precise

One more power: next-to-next-to-leading order (NNLO)

Even more complex [entangled infinities], more precise

Calculations are complex & done numerically

Nature isn't limited to perturbation theory

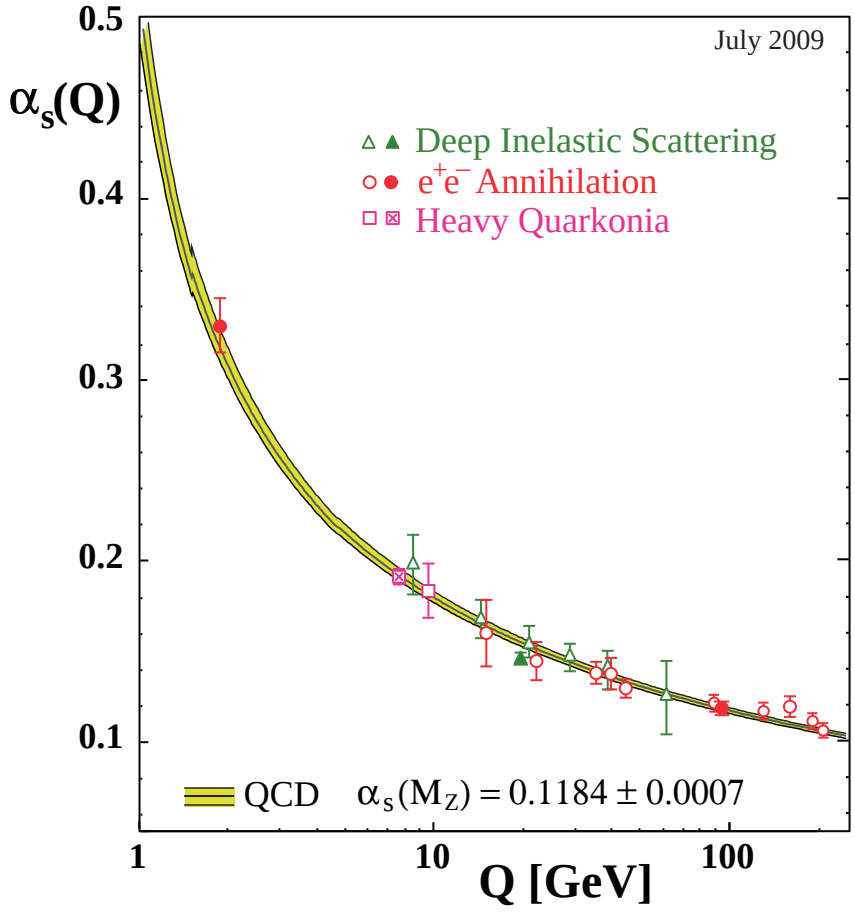
running of α_s in a theoretical calculation (yellow shaded band) and in physical processes at different characteristic scales from Bethke:2009jm

Confinement

Infrared slavery

No FREE quarks/gluons

Non-perturbative physics



Asymptotic freedom

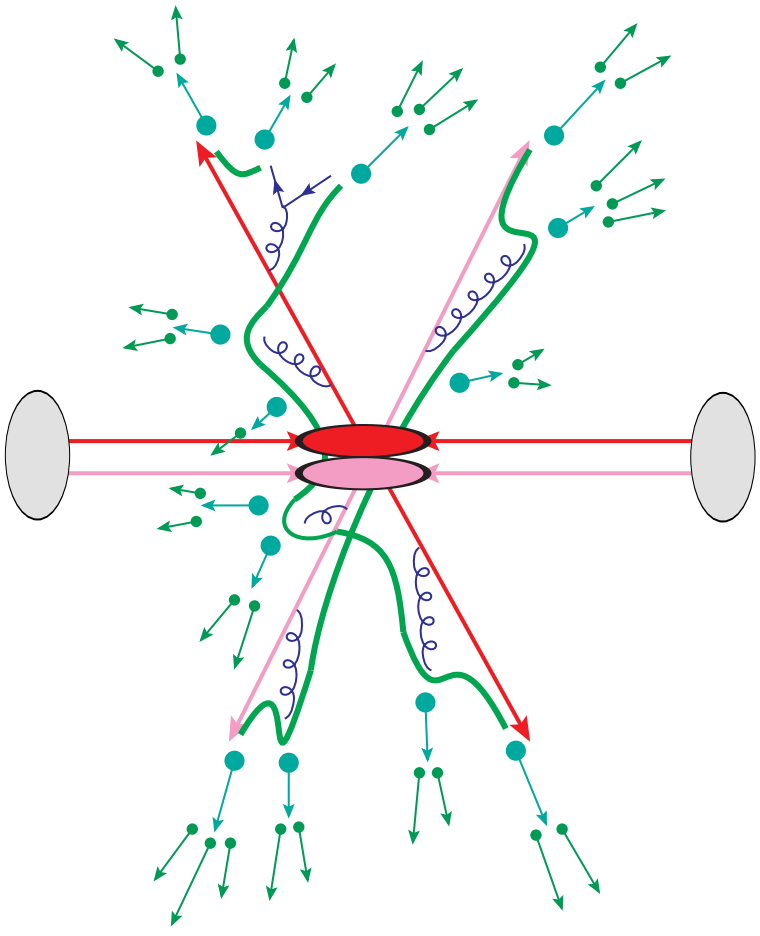
quarks/gluons FREE at short distances

Perturbation theory!

$M_{Higgs} \sim 125$ GeV

What is Pythia8?

A machine that generates realistic, high-energy particle collision event records using all the Standard Model physics we know and inspired models to fill in the rest.



Many diverse phenomenom

Short-distance cross section:
μ_r^H, μ_f^H, PDF^H, α_s^H
Parton shower:
μ_q^{PS}, μ_r^{PS}, μ_f^{PS}, μ_{cut}^{PS}, PDF^{PS}, α_s^{PS}
Multiple interactions:
μ_q^{MPI}, PDF^{MPI}, α_s^{MPI}...
String fragmentation:
$f(z)$, string p_T, tension
Beams:
Primordial k_T, remnants
Particle Decays:
BRs, MEs, BE correlations
...

What is it made of?

- $>$ 100,000 lines of code (C++)
- $\sim$ 1000 physics "rules"
- $\sim$ 1000 parameters (booleans, integers, real numbers)
- $>$ 40,000 lines of documentation
- $\sim$ 1000 files

- thousands of users worldwide
- 20,000+ citations to PYTHIA/JETSET manuals

Where does it come from?



The Pythia8 Collaboration

Christian Bierlich (Lund)

Naomi Cooke (Glasgow)

Nishita Desai (Tata)

Leif Gellersen (Lund)

Ilkka Helenius (Jyväskylä)

Philip Ilten (Cincinnati)

Leif Lönnblad (Lund)

Stephen Mrenna (FNAL)

Christian Preuss (ETH Zürich)

Torbjörn Sjöstrand (Lund)

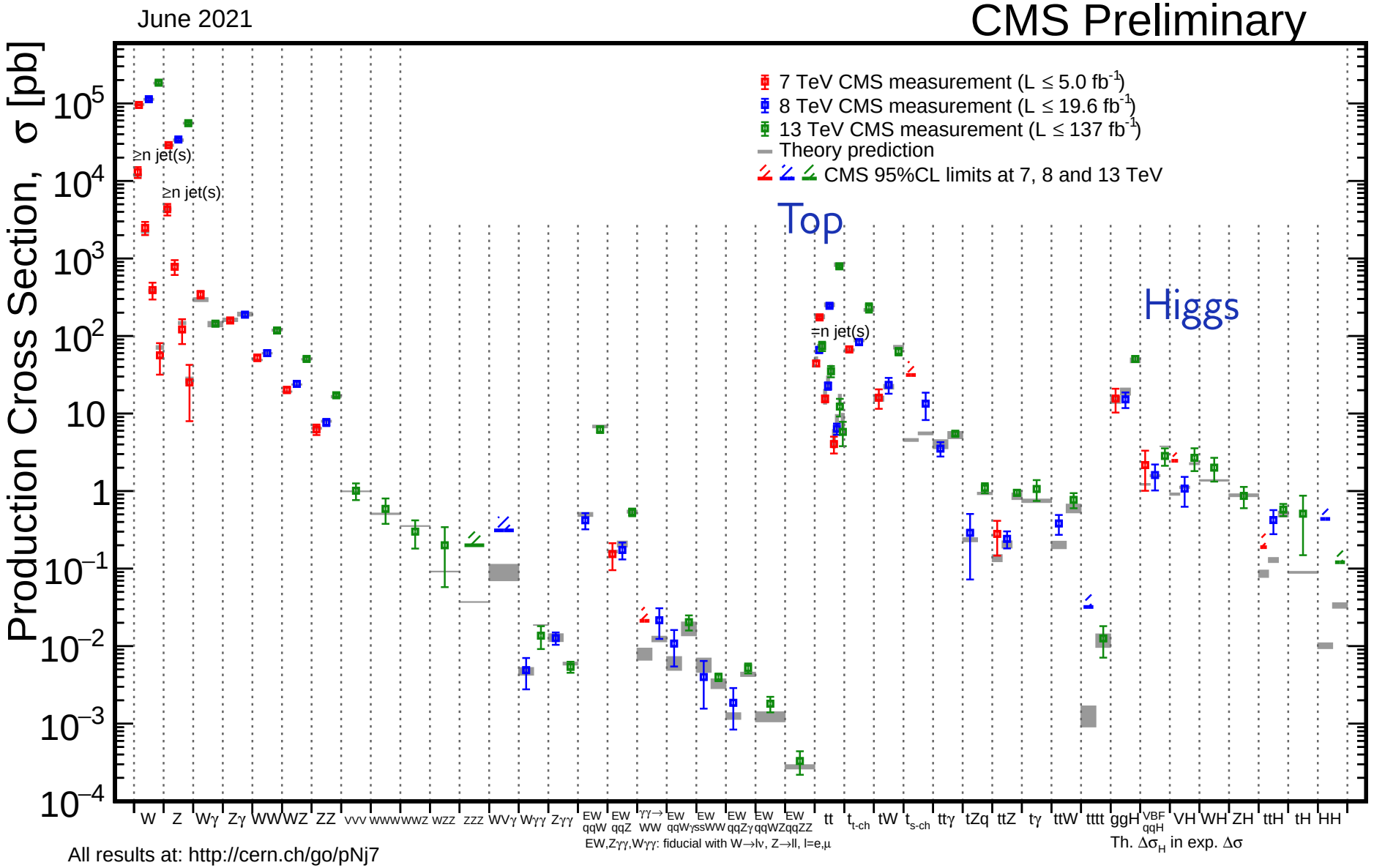
Peter Skands (Monash)

Marius Utheim (Jyväskylä)

Rob Verheyen (UC London)

What is it good for?

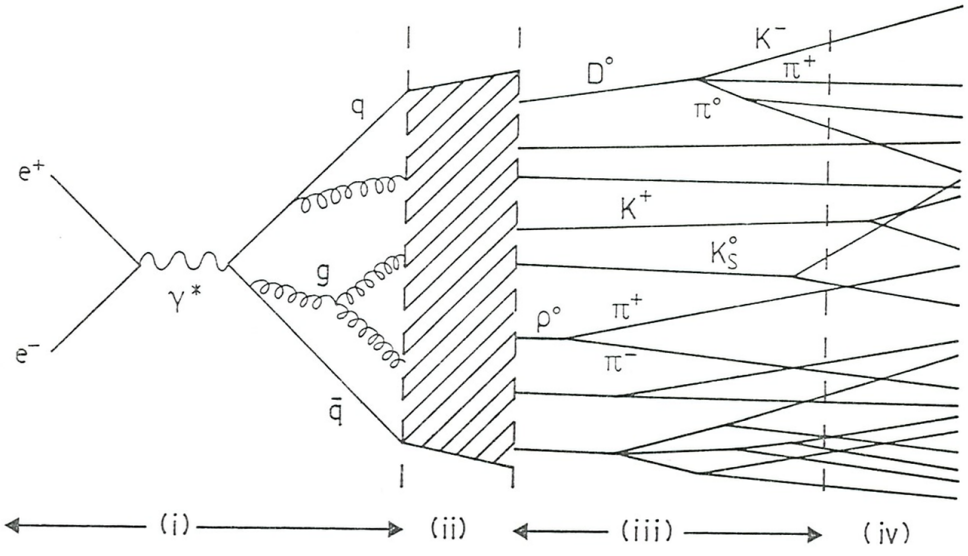
Theory vs. Data for the Standard Model



Origins of Pythia

Modern event generators were born at DESY (Hamburg), for the PETRA e^+e^- collider! (1978 – 86, 13 – 46 GeV)

- Combined (i) perturbative picture of hard processes, involving electroweak and strong interactions, with (ii) nonperturbative picture of hadronization, (iii) primary and (iv) secondary decays
- Provided "complete" events with parameters to be tuned to data. Used to study and understand different kinds of physics.



JETSET version 1 (November 1978)

```
      SUBROUTINE JETGEN(N)
      COMMON /JET/ K(100,2), P(100,5)
      COMMON /PAR/ PUD, PS1, SIGMA, CX2, EBEG, WFIN, IFLBEG
      COMMON /DATA1/ MESO(9,2), CMIX(6,2), PMAS(19)
      IFLSGN=(10-IFLBEG)/5
      W=2.*EBEG
      I=0
      IPD=0
C 1 FLAVOUR AND PT FOR FIRST QUARK
      IFL1=IABS(IFLBEG)
      PT1=SIGMA*SQRT(-ALOG(RANF(0)))
      PHI1=6.2832*RANF(0)
      PX1=PT1*COS(PHI1)
      PY1=PT1*SIN(PHI1)
  100 I=I+1
C 2 FLAVOUR AND PT FOR NEXT ANTIQUARK
      IFL2=1+INT(RANF(0)/PUD)
      PT2=SIGMA*SQRT(-ALOG(RANF(0)))
      PHI2=6.2832*RANF(0)
      PX2=PT2*COS(PHI2)
      PY2=PT2*SIN(PHI2)
C 3 MESON FORMED, SPIN ADDED AND FLAVOUR MIXED
      K(I,1)=MESO(3*(IFL1-1)+IFL2,IFLSGN)
      ISPIN=INT(PS1+RANF(0))
      K(I,2)=1+9*ISPIN+K(I,1)
      IF(K(I,1).LE.6) GOTO 110
      TMIX=RANF(0)
      KM=K(I,1)-6+3*ISPIN
      K(I,2)=8+9*ISPIN+INT(TMIX+CMIX(KM,1))+INT(TMIX+CMIX(KM,2)
C 4 MESON MASS FROM TABLE, PT FROM CONSTITUENTS
  110 P(I,5)=PMAS(K(I,2))
      P(I,1)=PX1+PX2
      P(I,2)=PY1+PY2
      PMTS=P(I,1)**2+P(I,2)**2+P(I,5)**2
C 5 RANDOM CHOICE OF X=(E+PZ)MESON/(E+PZ)AVAILABLE GIVES E AND
      X=RANF(0)
      IF(RANF(0).LT.CX2) X=1.-X**(1./3.)
      P(I,3)=(X*W-PMTS/(X*W))/2.
      P(I,4)=(X*W+PMTS/(X*W))/2.
C 6 IF UNSTABLE, DECAY CHAIN INTO STABLE PARTICLES
  120 IPD=IPD+1
      IF(K(IPD,2).GE.8) CALL DECAY(IPD,I)
      IF(IPD.LT.I.AND.I.LE.96) GOTO 120
C 7 FLAVOUR AND PT OF QUARK FORMED IN PAIR WITH ANTIQUARK ABOVE
      IFL1=IFL2
      PX1=-PX2
      PY1=-PY2
C 8 IF ENOUGH E+PZ LEFT, GO TO 2
      W=(1.-X)*W
      IF(W.GT.WFIN.AND.I.LE.95) GOTO 100
      N=I
      RETURN
      END
```

RANF

```
      SUBROUTINE DECAY(IPD,I)
      COMMON /JET/ K(100,2), P(100,5)
      COMMON /DATA1/ MESO(9,2), CMIX(6,2), PMAS(19)
      COMMON /DATA2/ IDCO(12), CBR(29), KDP(29,3)
      DIMENSION U(3), BE(3)
C 1 DECAY CHANNEL CHOICE, GIVES DECAY PRODUCTS
      TBR=RANF(0)
      IDC=IDCO(K(IPD,2)-7)
  100 IDC=IDC+1
      IF(TBR.GT.CBR(IDC)) GOTO 100
      ND=(59+KDP(IDC,3))/20
      DO 110 I1=I+1,I+ND
      K(I1,1)=-IPD
      K(I1,2)=KDP(IDC,I1-I)
  110 P(I1,5)=PMAS(K(I1,2))
C 2 IN THREE-PARTICLE DECAY CHOICE OF INVARIANT MASS OF PRODUCTS 2+3
      IF(ND.EQ.2) GOTO 130
      SA=(P(IPD,5)+P(I+1,5))**2
      SB=(P(IPD,5)-P(I+1,5))**2
      SC=(P(I+2,5)+P(I+3,5))**2
      SD=(P(I+2,5)-P(I+3,5))**2
      TDU=(SA-SD)*(SB-SC)/(4.*SQRT(SB*SC))
      IF(K(IPD,2).GE.11) TDU=SQRT(SB*SC)*TDU**3
  120 SX=SC+(SB-SC)*RANF(0)
      TDF=SQRT((SX-SA)*(SX-SB)*(SX-SC)*(SX-SD))/SX
      IF(K(IPD,2).GE.11) TDF=SX*TDF**3
      IF(RANF(0)*TDU.GT.TDF) GOTO 120
      P(100,5)=SQRT(SX)
C 3 TWO-PARTICLE DECAY IN CM, TWICE TO SIMULATE THREE-PARTICLE DECAY
  130 DO 160 IL=1,ND-1
      IO=(IL-1)*100-(IL-2)*IPD
      I1=I+IL
      I2=(ND-IL-1)*100-(ND-IL-2)*(I+IL+1)
      PA=SQRT((P(IO,5)**2-(P(I1,5)+P(I2,5))**2)*
     &(P(IO,5)**2-(P(I1,5)-P(I2,5))**2))/(2.*P(IO,5))
  140 U(3)=2.*RANF(0)-1.
      PHI=6.2832*RANF(0)
      U(1)=SQRT(1.-U(3)**2)*COS(PHI)
      U(2)=SQRT(1.-U(3)**2)*SIN(PHI)
      TDA=1.-(U(1)*P(IO,1)+U(2)*P(IO,2)+U(3)*P(IO,3))**2/
     &(P(IO,1)**2+P(IO,2)**2+P(IO,3)**2)
      IF(K(IPD,2).GE.11.AND.IL.EQ.2.AND.RANF(0).GT.TDA) GOTO 140
      DO 150 J=1,3
      P(I1,J)=PA*U(J)
  150 P(I2,J)=-PA*U(J)
      P(I1,4)=SQRT(PA**2+P(I1,5)**2)
  160 P(I2,4)=SQRT(PA**2+P(I2,5)**2)
C 4 DECAY PRODUCTS LORENTZ TRANSFORMED TO LAB SYSTEM
      DO 190 IL=ND-1,1,-1
      IO=(IL-1)*100-(IL-2)*IPD
      DO 170 J=1,3
  170 BE(J)=P(IO,J)/P(IO,4)
      GA=P(IO,4)/P(IO,5)
      DO 190 I1=I+IL,I+ND
      BEP=BE(1)*P(I1,1)+BE(2)*P(I1,2)+BE(3)*P(I1,3)
      DO 180 J=1,3
  180 P(I1,J)=P(I1,J)+GA*(GA/(1.+GA)*BEP+P(I1,4))*BE(J)
  190 P(I1,4)=GA*(P(I1,4)+BEP)
      I=I+ND
      RETURN
      END
```

```
      SUBROUTINE EDIT(N)
      COMMON /JET/ K(100,2), P(100,5)
      COMMON /EDPAR/ ITHROW, PZMIN, PMIN, THETA, PHI, BETA(3)
      REAL ROT(3,3), PR(3)
C 1 THROW AWAY NEUTRALS OR UNSTABLE OR WITH TOO LOW PZ OR P
      I1=0
      DO 110 I=1,N
      IF(ITHROW.GE.1.AND.K(I,2).GE.8) GOTO 110
      IF(ITHROW.GE.2.AND.K(I,2).GE.6) GOTO 110
      IF(ITHROW.GE.3.AND.K(I,2).EQ.1) GOTO 110
      IF(P(I,3).LT.PZMIN.OR.P(I,4)**2-P(I,5)**2.LT.PMIN**2) GOTO 110
      I1=I1+1
      K(I1,1)=IDIM(K(I,1),0)
      K(I1,2)=K(I,2)
      DO 100 J=1,5
  100 P(I1,J)=P(I,J)
  110 CONTINUE
      N=I1
C 2 ROTATE TO GIVE JET PRODUCED IN DIRECTION THETA, PHI
      IF(THETA.LT.1E-4) GOTO 140
      ROT(1,1)=COS(THETA)*COS(PHI)
      ROT(1,2)=-SIN(PHI)
      ROT(1,3)=SIN(THETA)*COS(PHI)
      ROT(2,1)=COS(THETA)*SIN(PHI)
      ROT(2,2)=COS(PHI)
      ROT(2,3)=SIN(THETA)*SIN(PHI)
      ROT(3,1)=-SIN(THETA)
      ROT(3,2)=0.
      ROT(3,3)=COS(THETA)
      DO 130 I=1,N
      DO 120 J=1,3
  120 PR(J)=P(I,J)
      DO 130 J=1,3
  130 P(I,J)=ROT(J,1)*PR(1)+ROT(J,2)*PR(2)+ROT(J,3)*PR(3)
C 3 OVERALL LORENTZ BOOST GIVEN BY BETA VECTOR
  140 IF(BETA(1)**2+BETA(2)**2+BETA(3)**2.LT.1E-8) RETURN
      GA=1./SQRT(1.-BETA(1)**2-BETA(2)**2-BETA(3)**2)
      DO 160 I=1,N
      BEP=BETA(1)*P(I,1)+BETA(2)*P(I,2)+BETA(3)*P(I,3)
      DO 150 J=1,3
  150 P(I,J)=P(I,J)+GA*(GA/(1.+GA)*BEP+P(I,4))*BETA(J)
  160 P(I,4)=GA*(P(I,4)+BEP)
      RETURN
      END
```

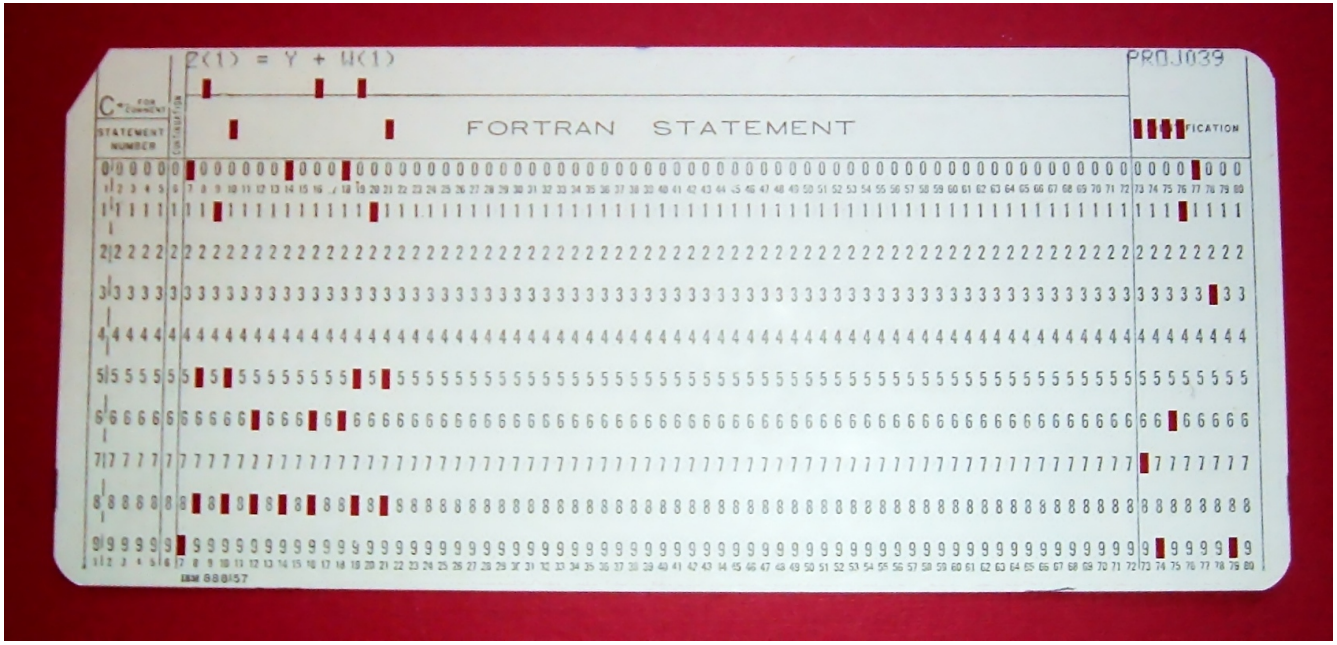
```
      SUBROUTINE LIST(N)
      COMMON /JET/ K(100,2), P(100,5)
      COMMON /DATA3/ CHA1(9), CHA2(19), CHA3(2)
      WRITE(6,110)
      DO 100 I=1,N
      IF(K(I,1).GT.0) C1=CHA1(K(I,1))
      IF(K(I,1).LE.0) IC1=-K(I,1)
      C2=CHA2(K(I,2))
      C3=CHA3((47-K(I,2))/20)
      IF(K(I,1).GT.0) WRITE(6,120) I, C1, C2, C3, (P(I,J), J=1,5)
  100 IF(K(I,1).LE.0) WRITE(6,130) I, IC1, C2, C3, (P(I,J), J=1,5)
      RETURN
  110 FORMAT(////T11,'I',T17,'ORI',T24,'PART',T32,'STAB',
     &T44,'PX',T56,'PY',T68,'PZ',T80,'E',T92,'M'/)
  120 FORMAT(10X,I2,4X,A2,1X,2(4X,A4),5(4X,F8.1))
  130 FORMAT(10X,I2,4X,1X,I2,2(4X,A4),5(4X,F8.1))
      END
```

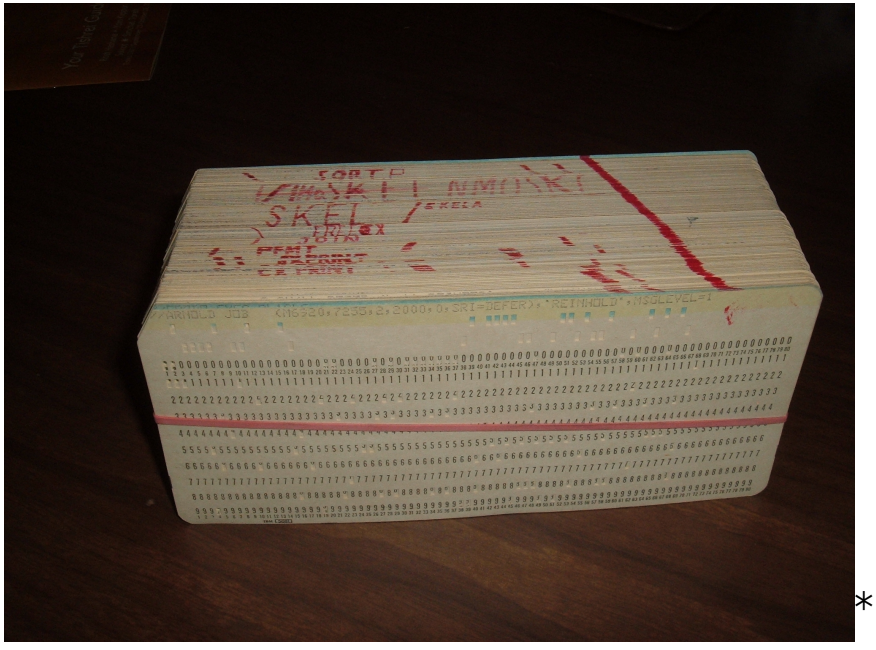
```
      BLOCK DATA
      COMMON /PAR/ PUD, PS1, SIGMA, CX2, EBEG, WFIN, IFLBEG
      COMMON /EDPAR/ ITHROW, PZMIN, PMIN, THETA, PHI, BETA(3)
      COMMON /DATA1/ MESO(9,2), CMIX(6,2), PMAS(19)
      COMMON /DATA2/ IDCO(12), CBR(29), KDP(29,3)
      COMMON /DATA3/ CHA1(9), CHA2(19), CHA3(2)
      DATA PUD/0.4/, PS1/0.5/, SIGMA/350./, CX2/0.77/,
     &EBEG/10000./, WFIN/100./, IFLBEG/1/
      DATA ITHROW/1/, PZMIN/0./, PMIN/0./, THETA,PHI,BETA/5*0./
      DATA MESO/7,1,3,2,8,5,4,6,9,7,2,4,1,8,6,3,5,9/
      DATA CMIX/2*0.5,1.,2*0.5,1.,2*0.25,0.5,2*0.,1./
      DATA PMAS/0.,2*139.6,2*493.7,2*497.7,135.,548.8,957.6,
     &2*765.9,2*892.2,2*896.3,770.2,782.6,1019.6/
      DATA IDCO/0,1,6,11,12,13,15,17,19,21,22,25/
      DATA CBR/1.,0.381,0.681,0.918,0.969,1.,0.426,0.662,0.959,
     &0.980,1.,1.,1.,0.667,1.,0.667,1.,0.667,1.,0.667,1.,1.,
     &0.899,0.987,1.,0.486,0.837,0.984,1./
      DATA KDP/1,1,8,2,1,1,2,8,1,1,1,2,3,6,4,7,5,4,6,5,7,2,2,
     &1,2,4,6,2,1,1,1,8,3,2,1,3,8,17,18,1,8,8,2,8,3,8,3,8,2,8,
     &3,3,8,3,5,7,3,9,0,0,8,8,3,8,9,9,14*0,8,4*0,8,0/
      DATA CHA1/'UD','DU','US','SU','DS','SD','UU','DD','SS'/
      DATA CHA2/'GAMM','PI+','PI-','K+','K-','KO','KBO','PIO','ETA',
     &'ETAP','RHO+','RHO-','K*+','K*-','K*O','KB*O','RHOO','OMEG','PHI'/
      DATA CHA3/'    ','STAB'/
      END
```

≈ 200 punched cards

Fortran code

(Sjostrand/Soderberg)



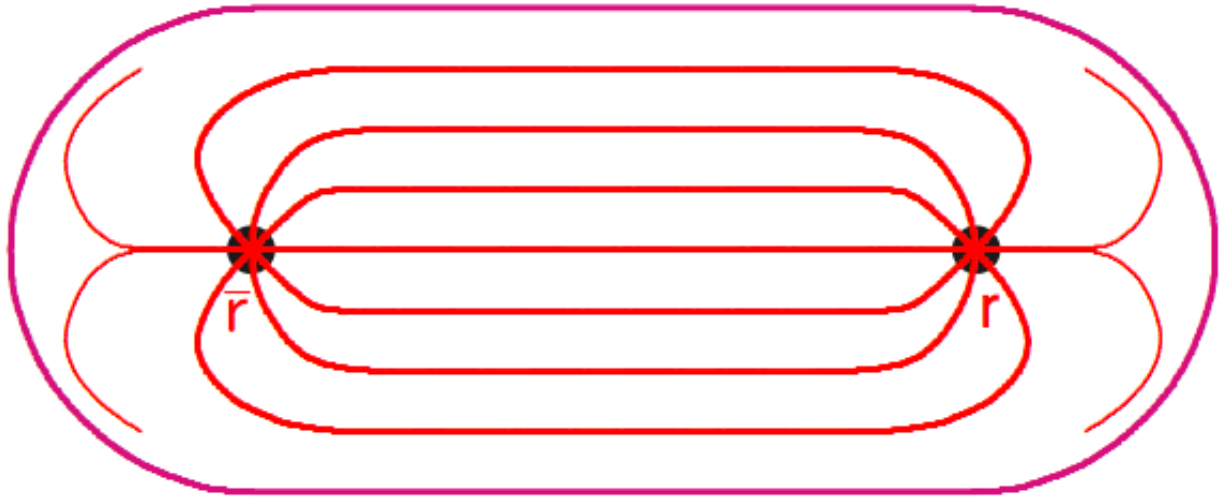


*

*https://en.wikipedia.org/wiki/Punched_card

The Lund String Model: Key component

In QCD, for large charge separation, field lines seem to be compressed to tubelike region(s) $\Rightarrow$ **string(s)**



by self-interactions among soft gluons in the "vacuum".

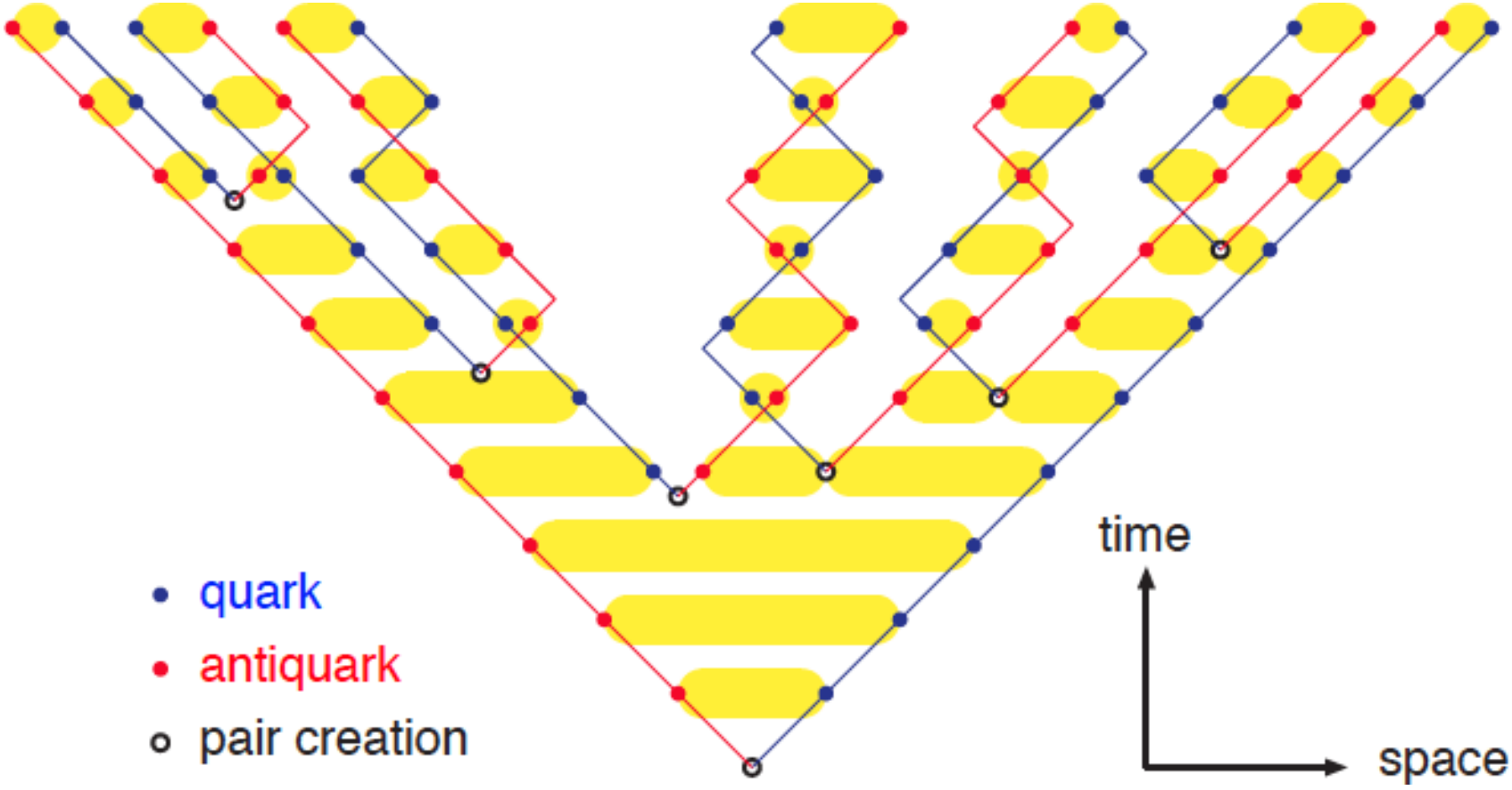
Gives linear confinement with string tension:

$F(r) \approx \text{const} = \kappa \approx 1\,\text{GeV/fm} \quad \Longleftrightarrow \quad V(r) \approx \kappa r$

String breaks into hadrons along its length,
with roughly uniform probability in rapidity,
by formation of new $q\bar{q}$ pairs that screen endpoint colors.

Lund String Model: 2 jets

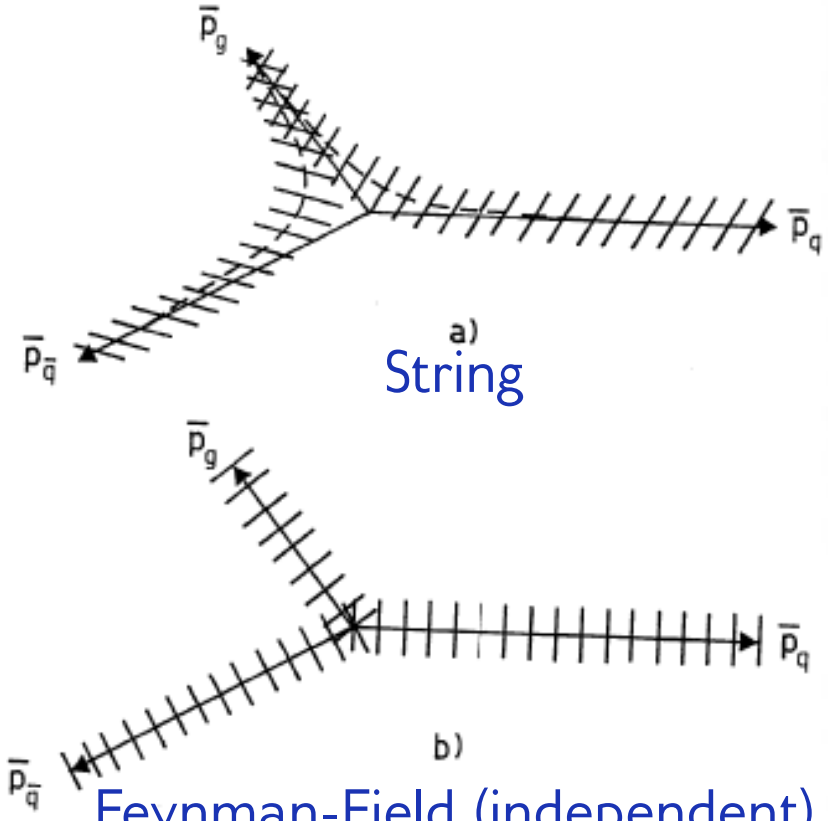
$e^+e^- \to q\bar{q}$



Consistent description of 2 jet topologies

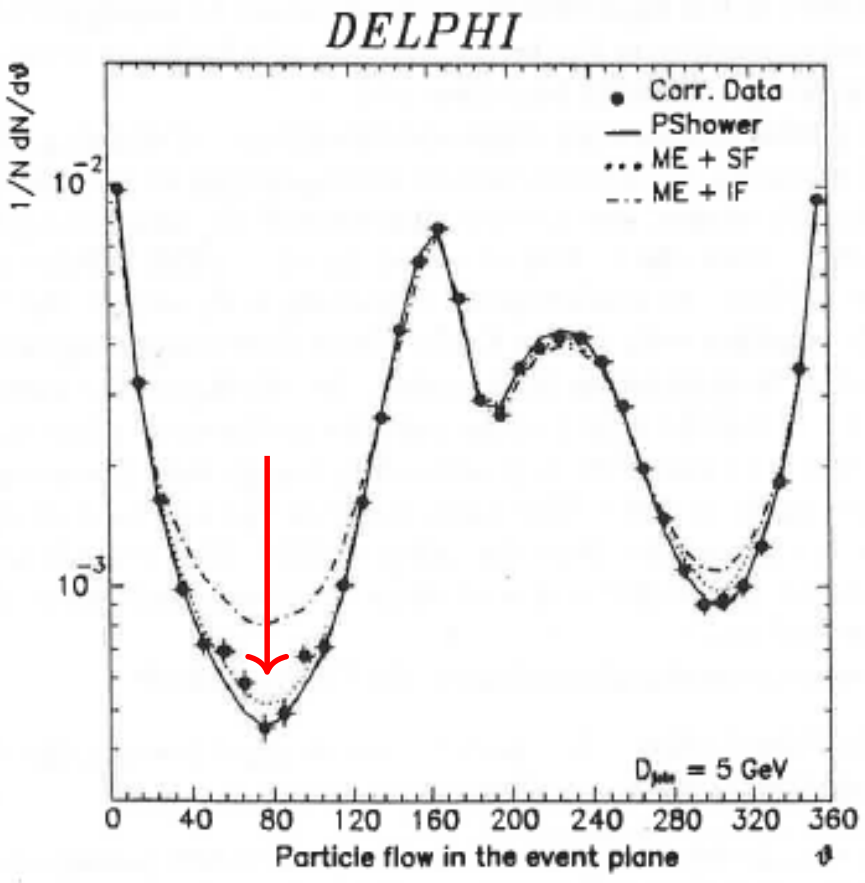
Lund String Model: 3 jets

No new parameters introduced for gluon jets!



String

Feynman-Field (independent)



Perturbation Theory: the Hard Process

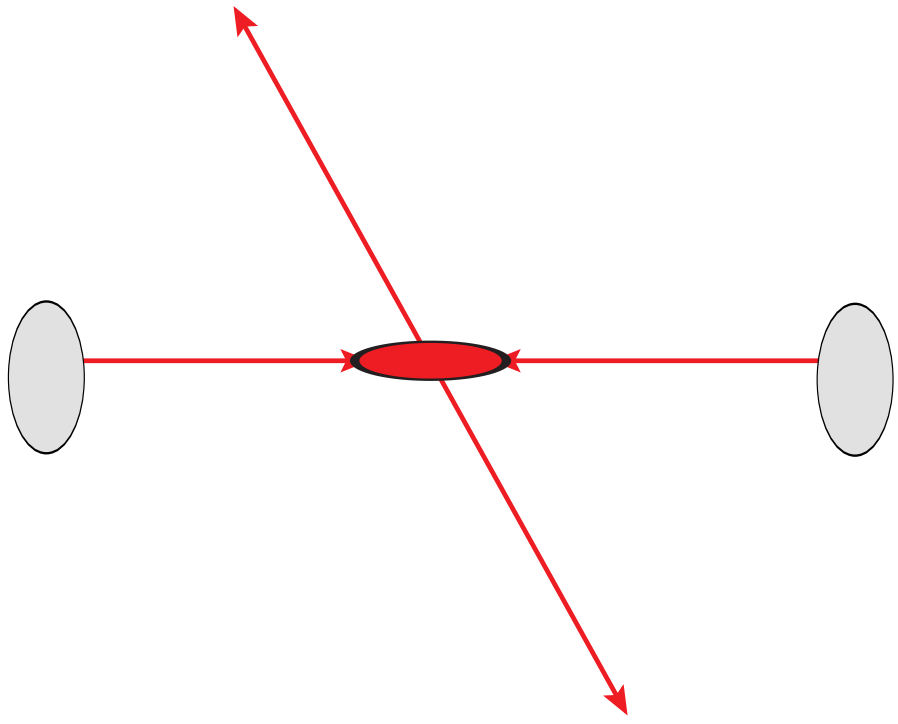
Long-distance and short-distance physics factorize.
(low-energy) *(high-energy)* *(we hope)*

$$\sigma = \int d\sigma_{(ab\to X+N \text{ partons})}(\text{high energy}) \otimes f_{a\in A}(\{x\}_a, \text{high energy}) \otimes f_{b\in B}(\{x\}_b, \text{high energy}) \otimes \mathcal{D}(p_A, p_B, p_1, \ldots, p_N) + \textit{corrections}$$

◇ Use perturbation theory to calculate $d\sigma$ at high energy.
Make accurate (NLO, NNLO) to capture most of the dynamics.
Hope: Less impact of non-perturbative dynamics.

◇ f and $\mathcal{D}$ are empirical, but have known scaling between energies

$f(x_a, \mu_F)$



$f(x_b, \mu_F)$

Dressing the Event: Soft-Collinear Radiation

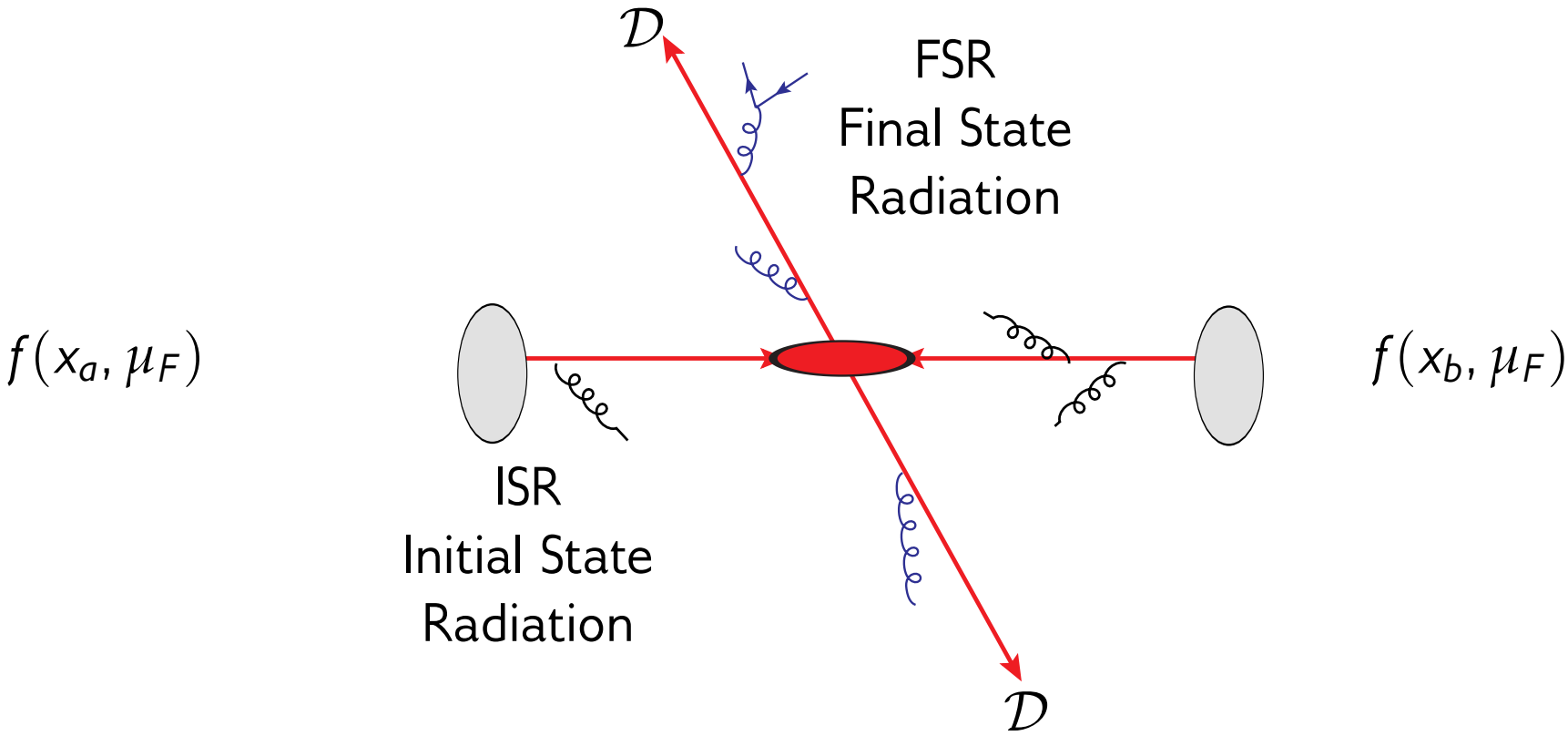
For each pair (a,b) of color-anticolor charges A and B in X, the cross sections for $X+1$ partons, $\mathrm{d}\sigma^{(0)}_{X+1}$, is:

$$\mathrm{d}\sigma^{(0)}_{X+1} = g_s^2 \left(\mathcal{N}_{AB\to acb} \frac{\mathrm{d}E_c}{E_c} \left(\frac{\mathrm{d}\theta_{ac}}{\theta_{ac}} + \frac{\mathrm{d}\theta_{bc}}{\theta_{bc}} \right) \; + \text{ less singular terms} \right) \mathrm{d}\sigma^{(0)}_X$$

When restricted to the phase-space region above some infrared cutoff scale μ_{IR}:

$$\sigma^{(0)}_{X+n} = \alpha_s^n \left(\ln^{2n} + \ln^{2n-1} + \ln^{2n-2} + \ldots + \ln \; + \mathcal{R} \right)$$

These logarithms are calculated using a Monte Carlo technique known as the parton shower.


$\mathcal{D}$
FSR
Final State
Radiation
$f(x_a, \mu_F)$
$f(x_b, \mu_F)$
ISR
Initial State
Radiation
$\mathcal{D}$

Matrix Elements and Parton Showers

2 types of perturbation theory

2 types of expansions that can be systematically corrected/improved

How do we combine them?

Complementary strengths:

- MEs describe well-separated, large-energy jets
- PSs describe soft/collinear structure inside jets

Higher precision demands ME corrections to PS or vice versa

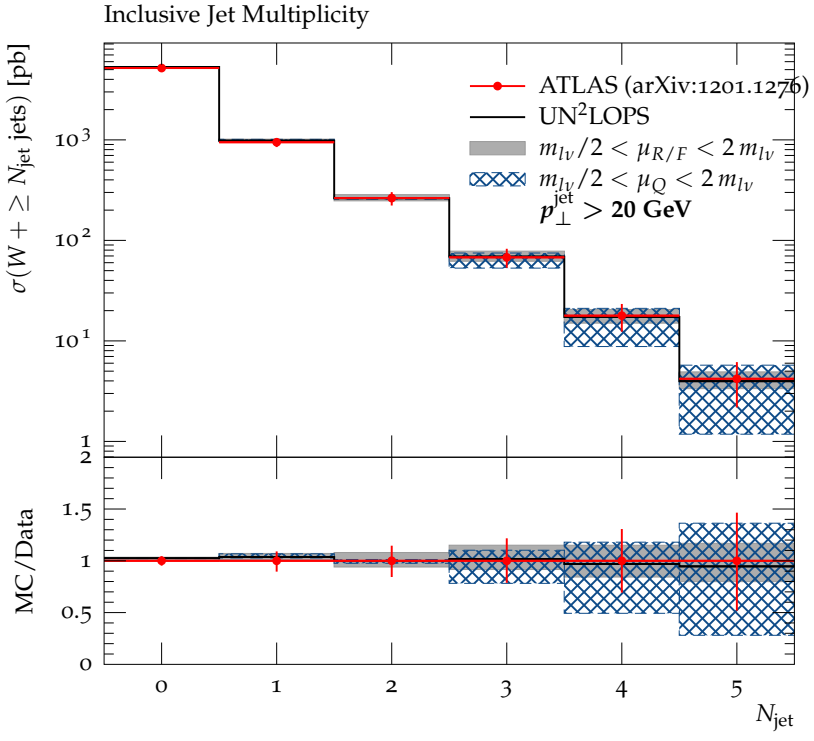
Very active field of research; many approaches

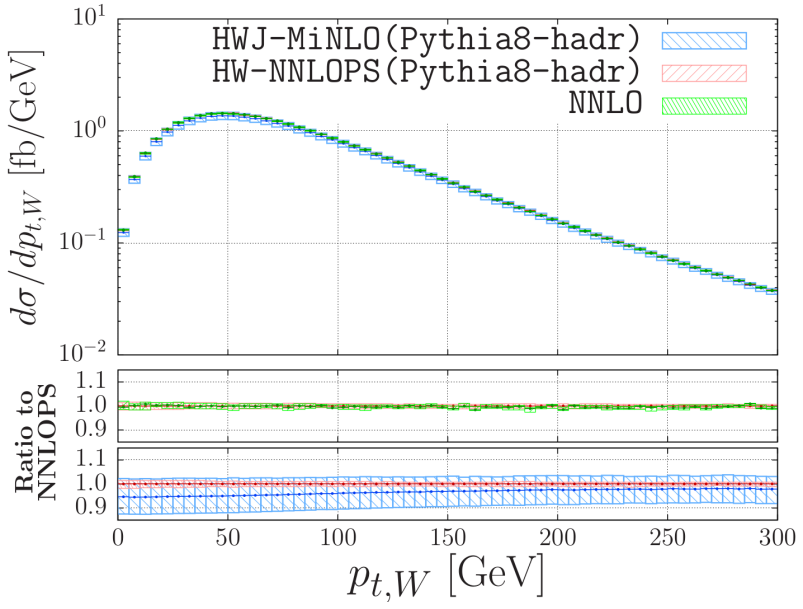
Landmark: NNLO+PS matching

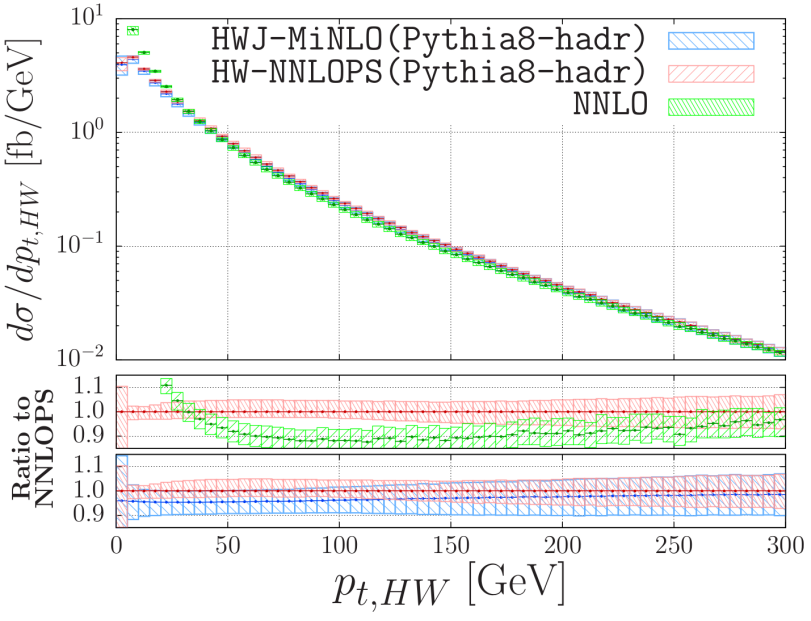
JHEP 1310 (2013) 222, arXiv:1405.3607, arXiv:1603.01620

- ✓NNLO for incl. cross-section
- ✓NLO+PS for 1-parton observables
- ✓LO+PS for 2-parton observables

Two schemes: NNLOPS/MINLO and UN^{2}LOPS

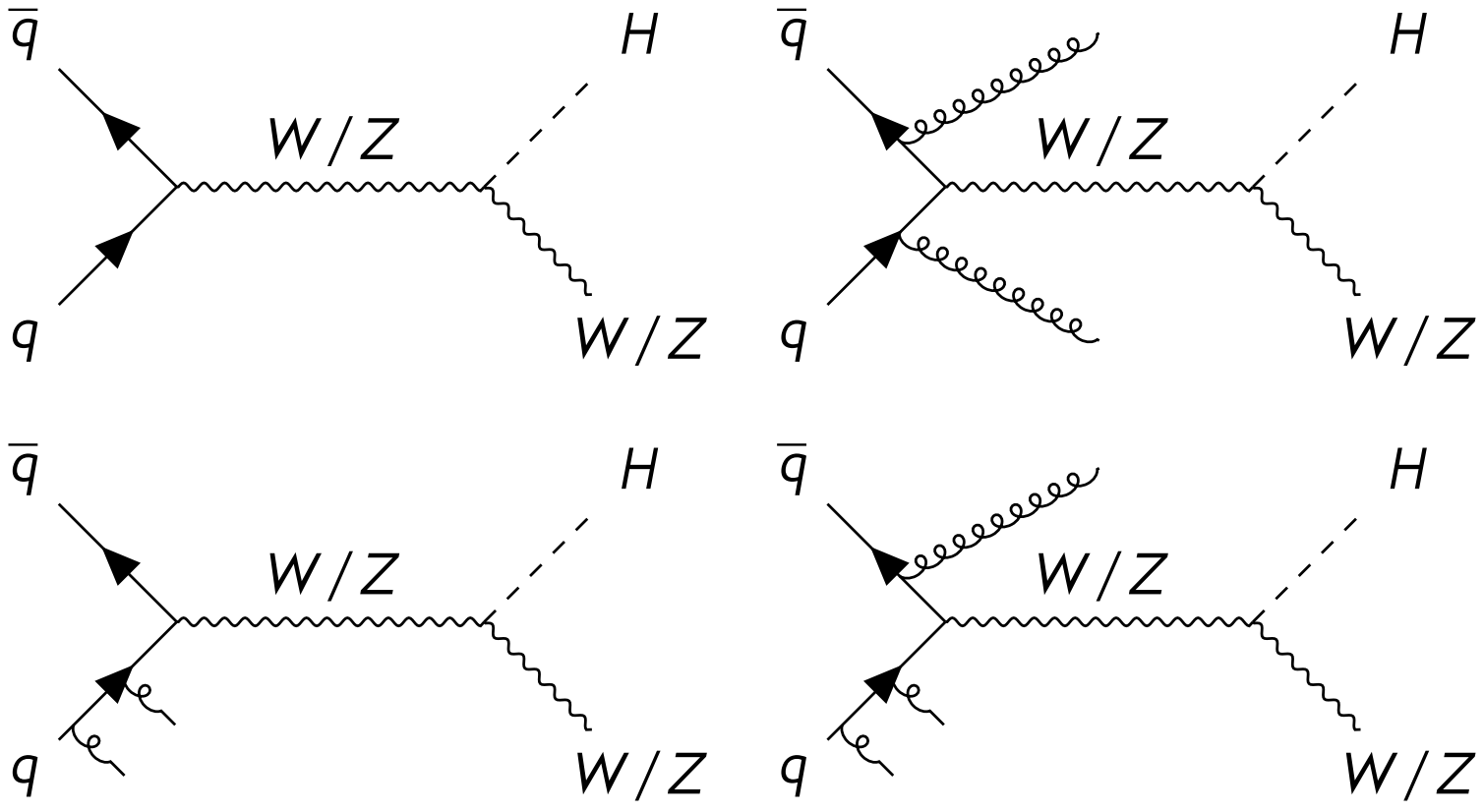






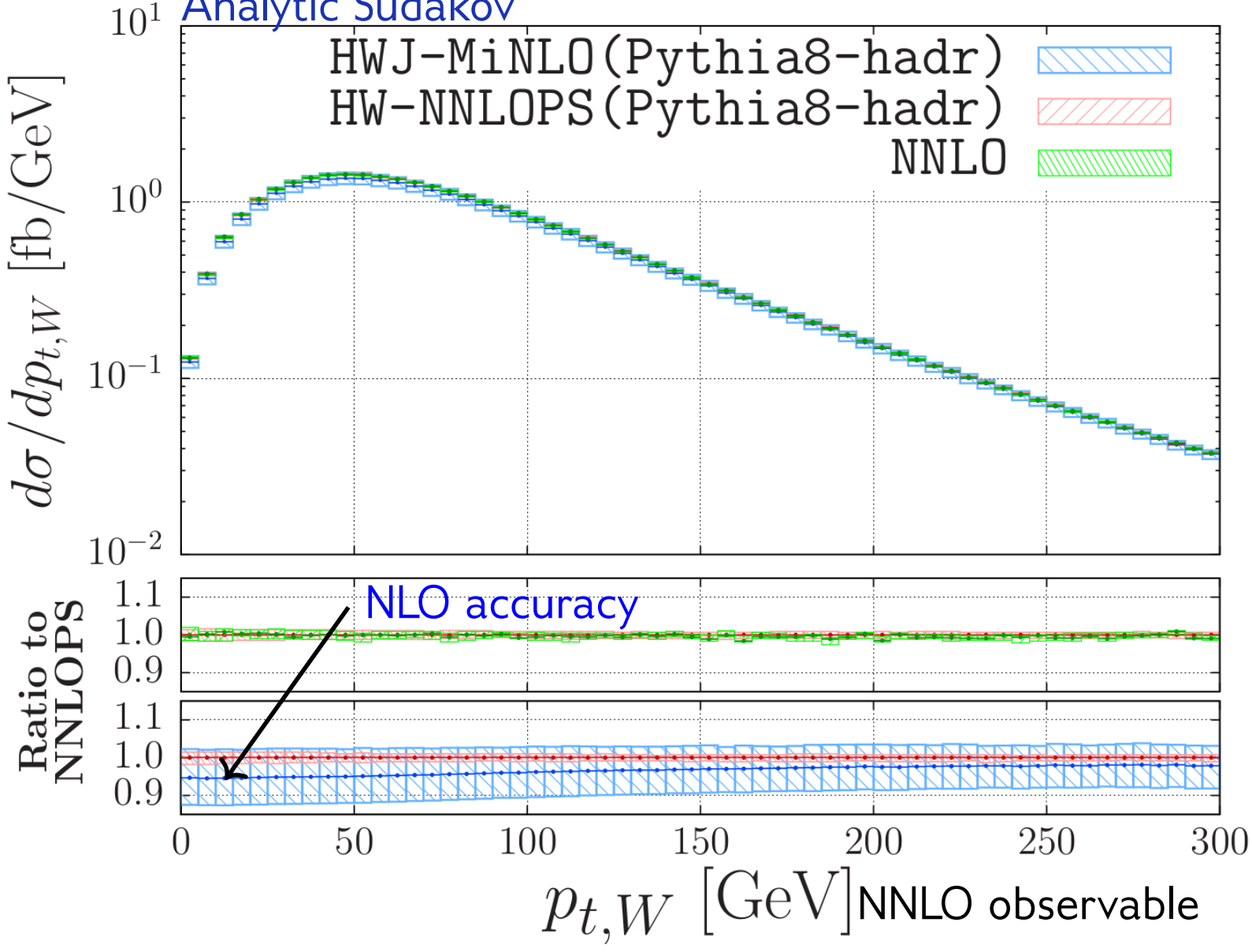
Some of the pieces

Feynman diagrams $\sim \sqrt{\text{probability}}$

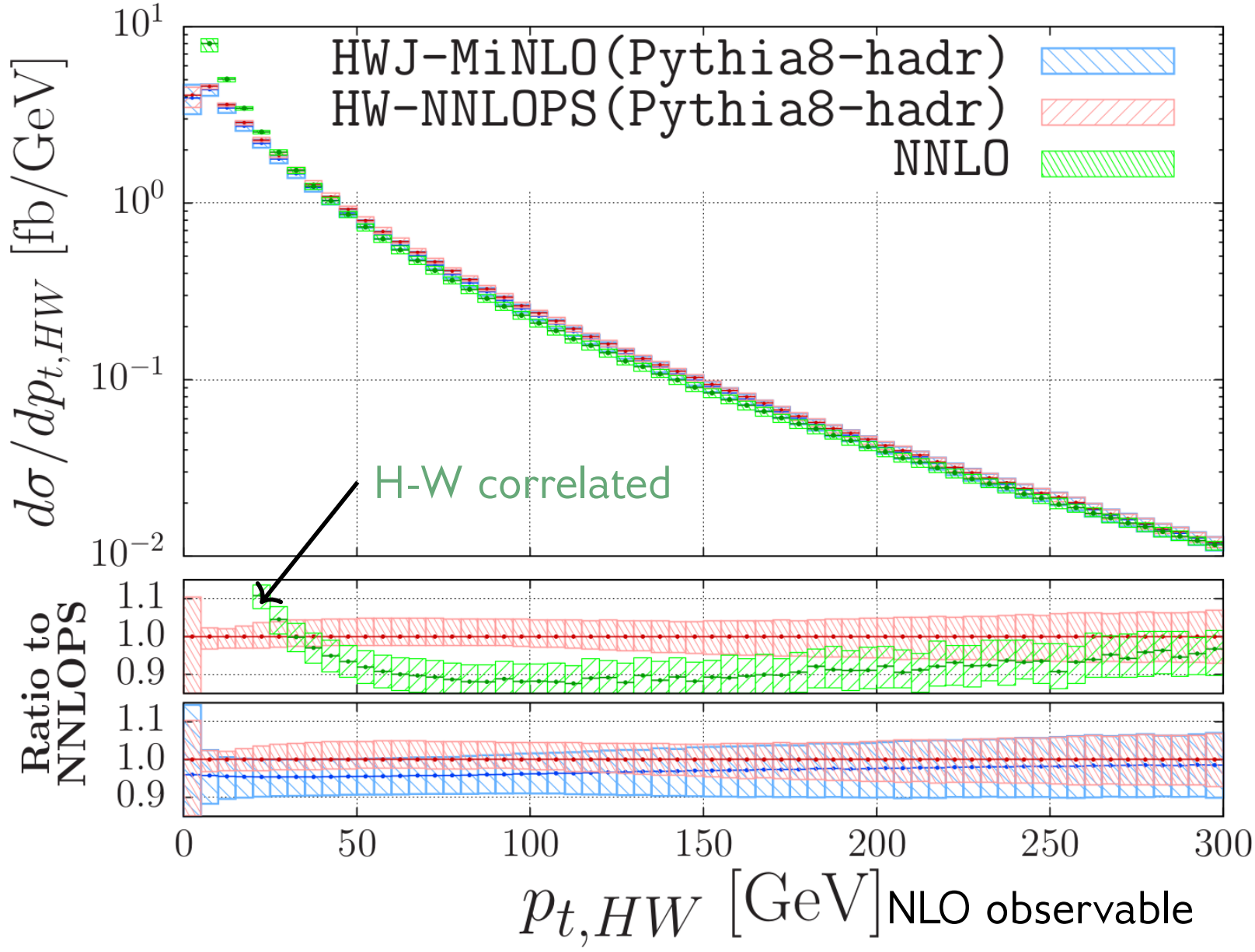


✓NNLO for incl. cross-section

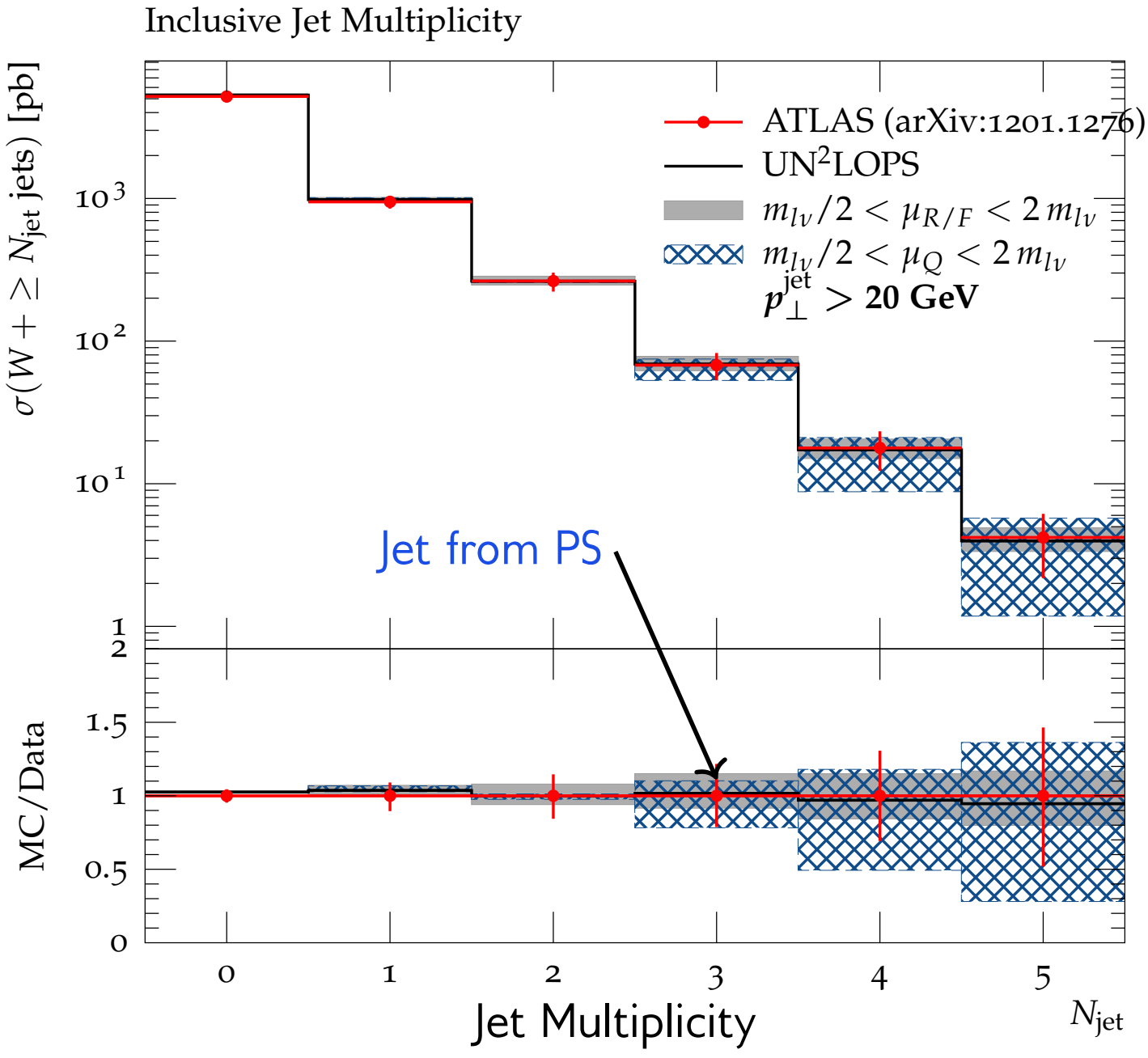
Analytic Sudakov



✓NLO+PS for 1-parton observables



✓LO+PS for 2-parton observables (*W*+jets)



The Bottom Line

High Level of sophistication in these calculations

Numerical methods an interesting scientific computing topic (how to identify subtracted infinities, calculate finite part, etc.)

More People work on the theoretically clean(er) aspects of perturbation theory

ME and PS are only a part of the story

Other phenomenon must be understand/controlled/subtracted to make a comparison to these predictions

Besides Hadronization (quarks/gluon $\rightarrow$ hadrons), other types of semi- or non-perturbative physics must be modelled

Multi-parton Interactions

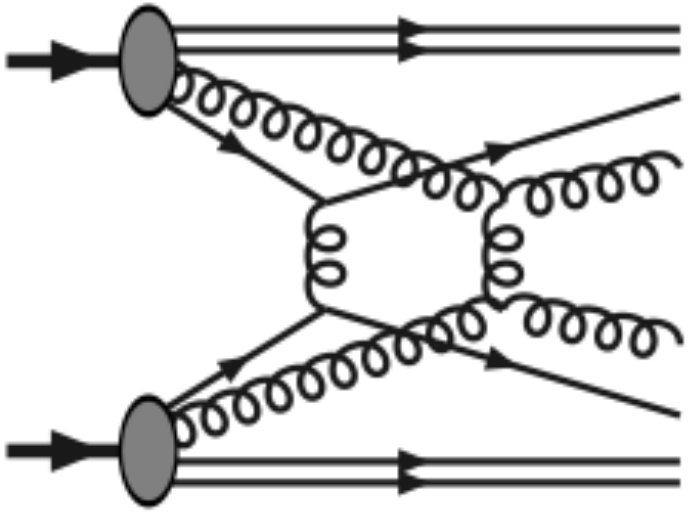
A bit of theory, a bit of model building

$$\sigma_{\mathrm{int}} = \iiint_{\mathrm{cutoff}} \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}p_\perp^2\; f_1(x_1, p_\perp^2)\, f_2(x_2, p_\perp^2)\, \frac{\mathrm{d}\hat{\sigma}}{\mathrm{d}p_\perp^2} = \langle n \rangle_{\mathrm{MPI}}\, \sigma_{\mathrm{inel}}$$

cutoff corresponds to color screening at small $p_\perp$

(proton finite size cuts off gluon wavelength)

$\sigma_{\mathrm{int}} > \sigma_{\mathrm{inel}}$ Crazy? or Just Counting?



Many parton-parton interactions per pp event: MPI.

Most have small $p_\perp$, $\sim$ 2 GeV
$\Rightarrow$ not visible as separate jets,
but contribute to event activity.

Solid evidence exists for MPIs setting event structure.

Multiparton interactions modeling

Simple → Complicated

1. Regularise cross section with $p_{\perp 0}$ as free parameter

$$\frac{\mathrm{d}\hat{\sigma}}{\mathrm{d}p_\perp^2} \propto \frac{\alpha_s^2(p_\perp^2)}{p_\perp^4} \to \frac{\alpha_s^2(p_{\perp 0}^2 + p_\perp^2)}{(p_{\perp 0}^2 + p_\perp^2)^2}$$

2. with energy dependence

$$p_{\perp 0}(E_{\mathrm{CM}}) = p_{\perp 0}^{\mathrm{ref}} \times \left(\frac{E_{\mathrm{CM}}}{E_{\mathrm{CM}}^{\mathrm{ref}}} \right)^{\epsilon}$$

3. Matter profile in impact-parameter space gives time-integrated overlap which determines level of activity:
4. Many colored partons produced close in space–time
   $\Rightarrow$ color rearrangement; reduction of total string length
   $\Rightarrow$ steeper $\langle p_\perp \rangle (n_{\mathrm{ch}})$

A large number of parameters need to be fit (tuned) to data to make predictions

Do we just over-fit data?

Actual parameters $<$ # Apparent parameters

Some parameters are just necessary choices (cutoffs, PDF, *etc*)

Sensitive parameters $<$ # Actual parameters

Information in differential data $\gg$ # Sensitive parameters

Different energies, different incoming beams

Perturbative, inclusive QCD calculations also have inputs: α_s, μ_R, μ_F, PDFs and make limited predictions

Minimal # parameters for LHC $\sim 15 - 20$

Depends on what question you are asking

Parameter Tuning

A Challenging Problem!

Divide and Conquer

- Start with data from electron-positron colliders
- Removes complication of the beams
  - No Initial State Radiation
  - No MPI
- Know collision energy E_{cm}
- $\cdots$ but limited range
- Most data @ $E_{\mathrm{cm}} = M_Z \sim 91\,\mathrm{GeV}$

Monash 2013 Tune Parameters

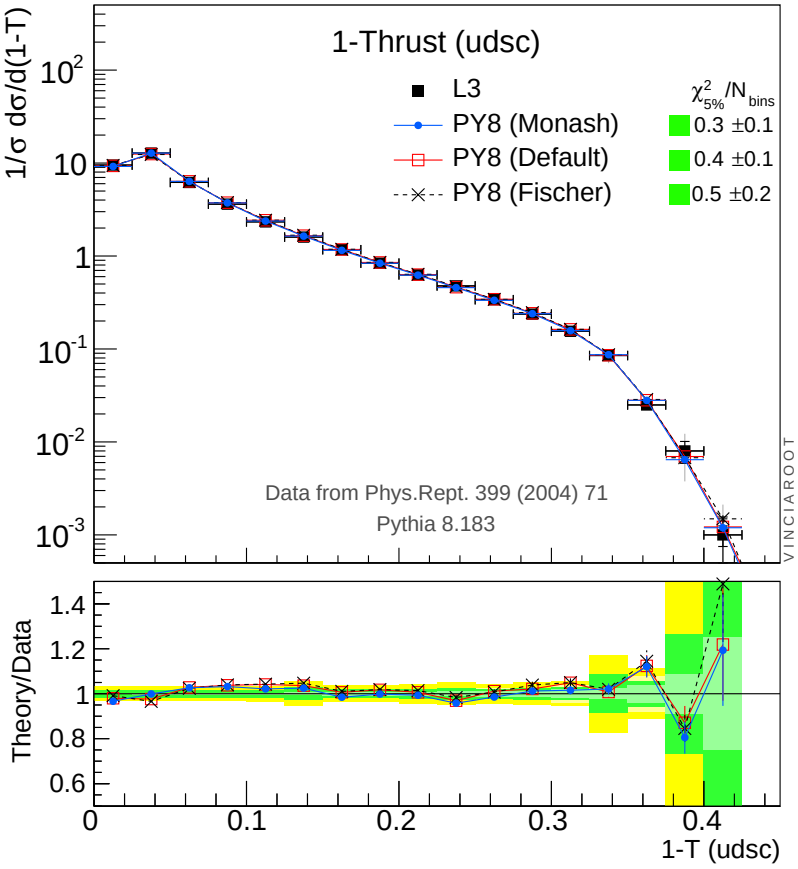
Final-state radiation (FSR) parameters.

FSR Parameters	Monash 13	(Default)	Comment
TimeShower:alphaSvalue	= 0.1365	= 0.1383	! Effective alphaS(mZ) value
TimeShower:alphaSorder	= 1	= 1	! Running order
TimeShower:alphaSuseCMW	= off	= off	! Translation from $\overline{\text{MS}}$ to CMW
TimeShower:pTmin	= 0.50	= 0.40	! Cutoff for QCD radiation
TimeShower:pTminChgQ	= 0.50	= 0.40	! Cutoff for QED radiation
TimeShower:phiPolAsym	= on	= on	! Asymmetric azimuth distributions

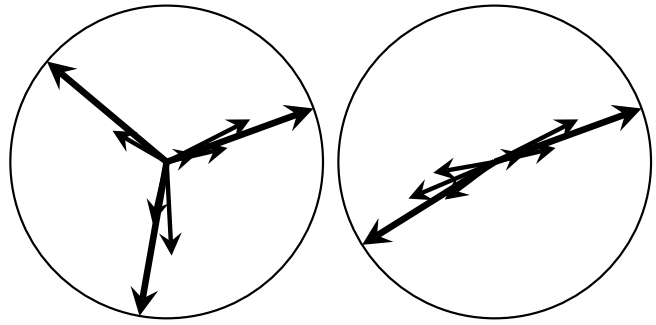
$$\text{FSR:} \quad \mu_R^2 = p_{\perp\text{evol}}^2 = z(1-z)Q^2 \ ,$$

with $Q^2 = p^2 - m_0^2$ the offshellness of the emitting parton (with on-shell mass m_0), and z the energy fraction appearing in the DGLAP splitting kernels, $P(z)$.

pTmin :: avoid Landau pole



Hadronic Z decays at $\sqrt{s} = 91.2\,\text{GeV}$. The Thrust distribution in light-flavor tagged events, compared with L3 data



$$\alpha_s(M_Z) = 0.1365 \neq 0.13$$

$$\text{CMW} \neq \overline{\text{MS}}$$

$$\Lambda_{\text{CMW}} = 1.6\Lambda_{\overline{\text{MS}}}$$

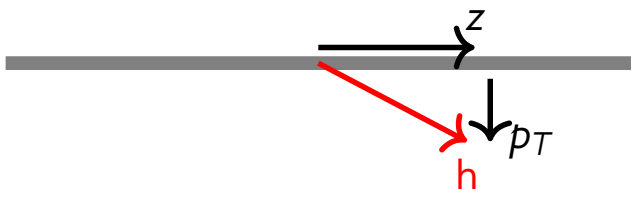
Choice of cutoff, hadronization parameters also impact prediction

String-breaking parameters.

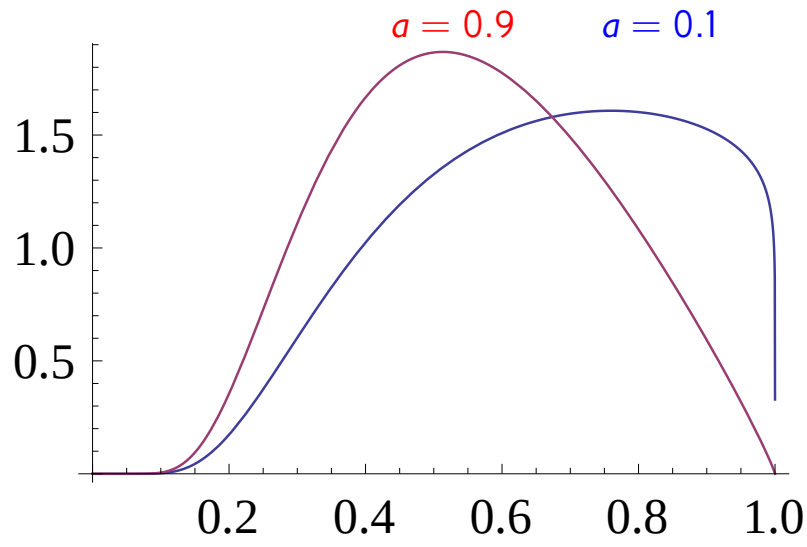
HAD Parameters	Monash 13	(Default)	Comment
# String breaks: pT and z distributions			
StringPT:sigma	= 0.335	= 0.304	! Soft pT in string breaks (in GeV)
StringPT:enhancedFraction	= 0.01	= 0.01	! Fraction of breakups with enhanced pT
StringPT:enhancedWidth	= 2.0	= 2.0	! Enhancement factor
StringZ:aLund	= 0.68	= 0.3	! Lund FF a (hard fragmentation supp)
StringZ:bLund	= 0.98	= 0.8	! Lund FF b (soft fragmentation supp)
StringZ:aExtraSquark	= 0.0	= 0.0	! Extra a when picking up an s quark
StringZ:aExtraDiquark	= 0.97	= 0.50	! Extra a when picking up a diquark
StringZ:rFactC	= 1.32	= 1.00	! Lund-Bowler c-quark parameter
StringZ:rFactB	= 0.855	= 0.67	! Lund-Bowler b-quark parameter
# Flavour composition: mesons			
StringFlav:ProbStoUD	= 0.217	= 0.19	! Strangeness-to-UD ratio
StringFlav:mesonUDvector	= 0.5	= 0.62	! Light-flavour vector suppression
StringFlav:mesonSvector	= 0.55	= 0.725	! Strange vector suppression
StringFlav:mesonCvector	= 0.88	= 1.06	! Charm vector suppression
StringFlav:mesonBvector	= 2.2	= 3.0	! Bottom vector suppression
StringFlav:etaSup	= 0.60	= 0.63	! Suppression of eta mesons
StringFlav:etaPrimeSup	= 0.12	= 0.12	! Suppression of eta' mesons
# Flavour composition: baryons			
StringFlav:probQQtoQ	= 0.081	= 0.09	! Diquark rate (for baryon production)
StringFlav:probSQtoQQ	= 0.915	= 1.000	! Strange-diquark suppression
StringFlav:probQQ1toQQ0	= 0.0275	= 0.027	! Vector diquark suppression
StringFlav:decupletSup	= 1.0	= 1.0	! Spin-3/2 baryon suppression
StringFlav:suppressLeadingB	= off	= off	! Optional leading-baryon suppression
StringFlav:popcornSpair	= 0.9	= 0.5	!
StringFlav:popcornSmeson	= 0.5	= 0.5	!

Lund symmetric fragmentation function:

$$f(z) \propto \frac{(1-z)^a}{z} \exp\left(\frac{-bm_\perp^2}{z}\right)$$

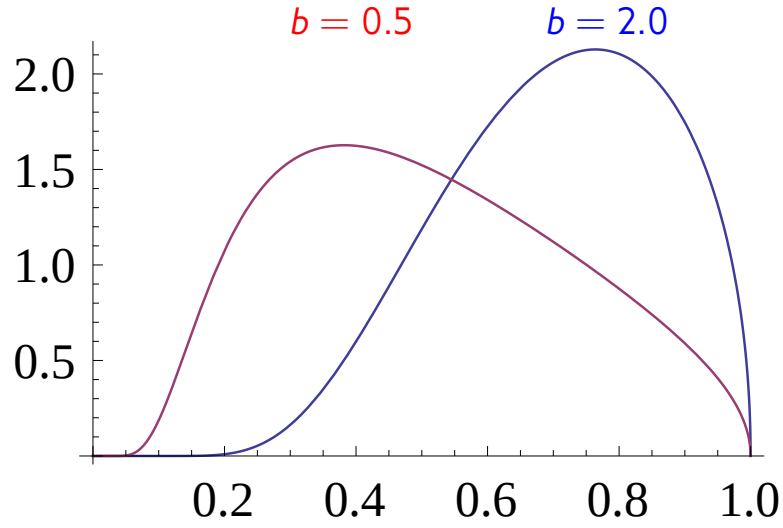


The a parameter

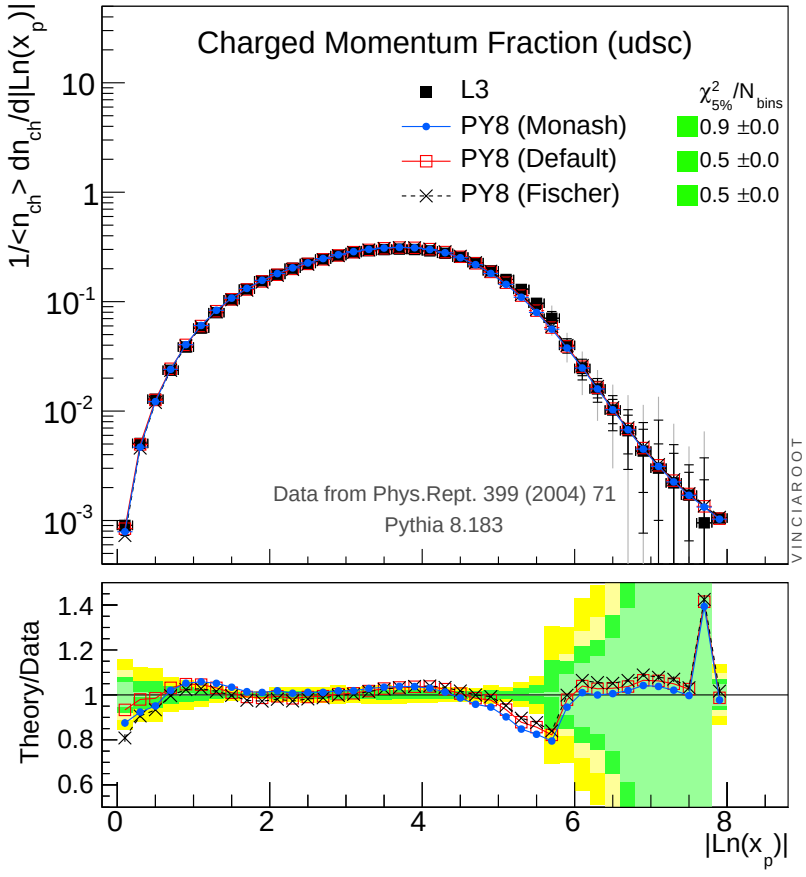


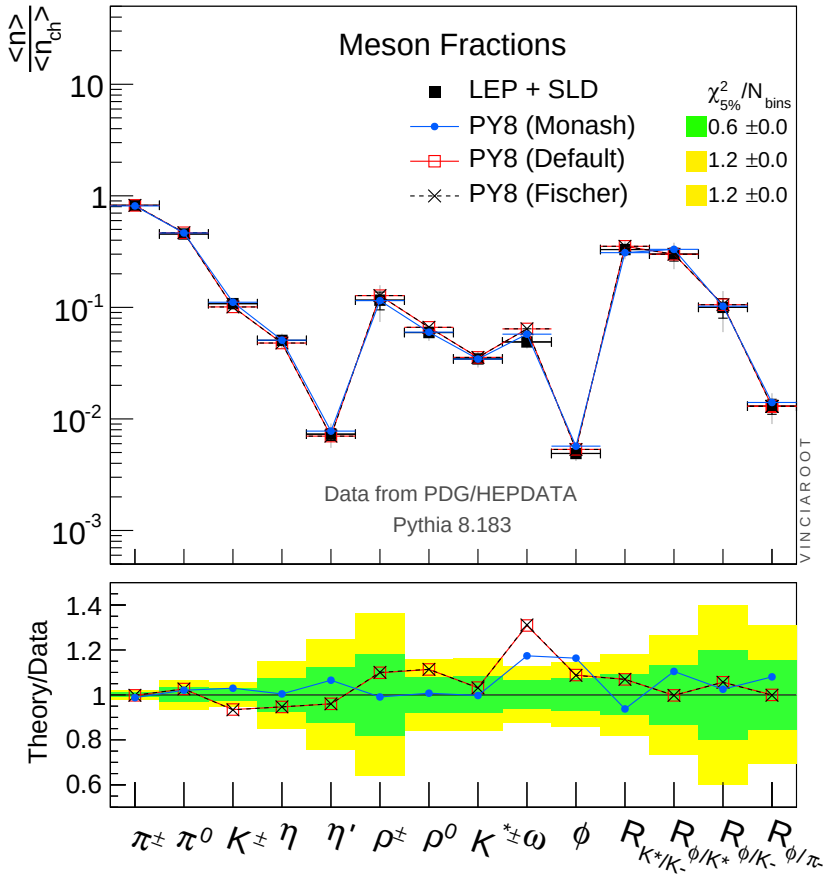
$b = 1\text{GeV}^{-2}$, $m_\perp = 1\text{GeV}$

The b parameter



$a = 0.5$, $m_\perp = 1\text{GeV}$






```
StringZ:aLund          =  0.68
StringZ:bLund          =  0.98

StringZ:aExtraDiquark =  0.97
StringZ:aExtraSquark  =  0.00
```

```
StringFlav:ProbStoUD     = 0.217
StringFlav:mesonUDvector = 0.5
StringFlav:mesonSvector  = 0.55
StringFlav:etaSup        = 0.60
StringFlav:etaPrimeSup   = 0.12
```

Universality Assumption

Final state effects (FSR, hadronization) are tuned in the relatively clean e^+e^- environment

- Problematic, because kinematics (energy range) is limited, e.g. not much $g \to Q\overline{Q}$ splitting

  Assume: Any differences in hadron collisions come from environment

- Color flow from initial to final state
- High density of colored partons

Must/should test this!

New parameters (models) for proton (nucleon) beams

Initial-state radiation (ISR) and primordial-k_T parameters.

ISR Parameters	Monash 13	(Default)	Comment
SpaceShower:alphaSvalue	= 0.1365	= 0.137	! Effective alphaS(mZ) value
SpaceShower:alphaSorder	= 1	= 1	! Running order
SpaceShower:alphaSuseCMW	= off	= off	! Translation from $\overline{\text{MS}}$ to CMW
SpaceShower:samePTasMPI	= off	= off	! ISR cutoff type
SpaceShower:pT0Ref	= 2.0	= 2.0	! ISR pT0 cutoff
SpaceShower:ecmRef	= 7000.0	= 1800.0	! ISR pT0 reference ECM scale
SpaceShower:ecmPow	= 0.0	= 0.0	! ISR pT0 scaling power
SpaceShower:rapidityOrder	= on	= on	! Approx coherence via y-ordering
SpaceShower:phiPolAsym	= on	= on	! Azimuth asymmetries from gluon pol
SpaceShower:phiIntAsym	= on	= on	! Azimuth asymmetries from interference
TimeShower:dampenBeamRecoil	= on	= on	! Recoil dampening in final-initial dipoles
BeamRemnants:primordialKTsoft	= 0.9	= 0.5	! Primordial kT for soft procs
BeamRemnants:primordialKThard	= 1.8	= 2.0	! Primordial kT for hard procs
BeamRemnants:halfScaleForKT	= 1.5	= 1.0	! Primordial kT soft/hard boundary
BeamRemnants:halfMassForKT	= 1.0	= 1.0	! Primordial kT soft/hard mass boundary

Plus parameters for MPI, partonic structure of proton, $\cdots$

Evolution of Pythia8 Tune in CMS Run2

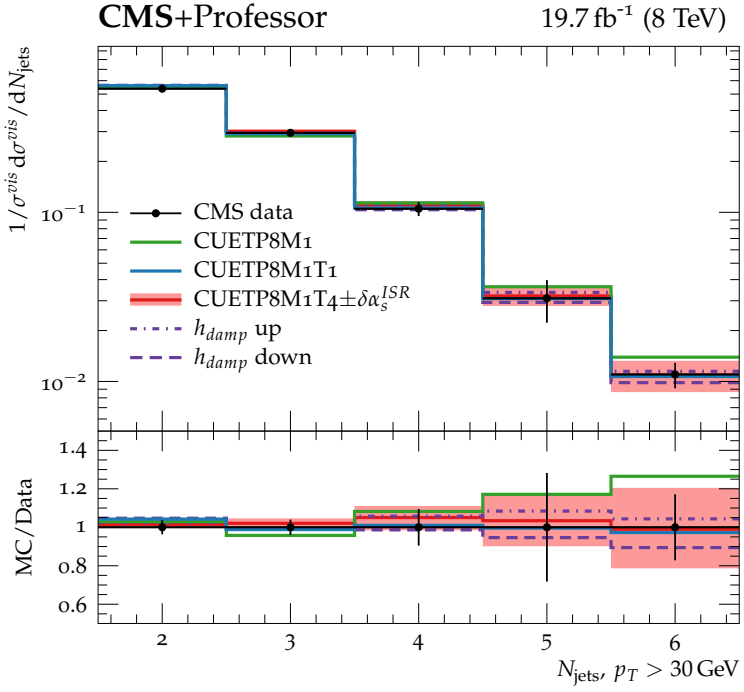
Monash (by-hand) tune was a good starting point, but new data came out afterwards at many different energies

- CMS used Tune CUETP8M1 for most samples (arXiv:1512.00815)
  - Re-tuning of UE parameters on top of Monash, α_S and other shower parameters left untouched
  - In particular this means α_S=0.1365 used for both ISR and FSR in the shower, despite using 0.118 in the ME for NLO samples and 0.130 for LO samples
- Tuning of shower parameters in particular revisited

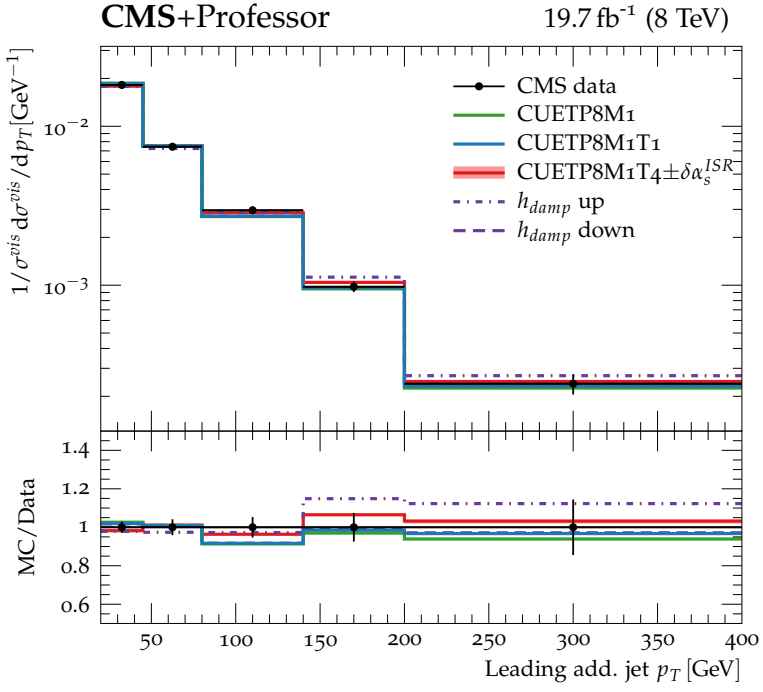
Pythia8 Shower Tuning with $t\bar{t}$ (CMS-PAS-TOP-16-021)

- Differences observed between generators, and sensitivity to shower α_s in $t\bar{t}$ production in kinematics/multiplicity of additional jets
- Retune PS ISR α_s and ME matching parameter using POWHEG+Pythia $t\bar{t}$ vs dilepton+jets data, yields (much) lower value of $\alpha_s^{ISR} = 0.1108$

Njets (p_T >30 GeV)



Lead jet p_T



What is theoretical uncertainty?

Long standing differences btw amc@nlo and pwg

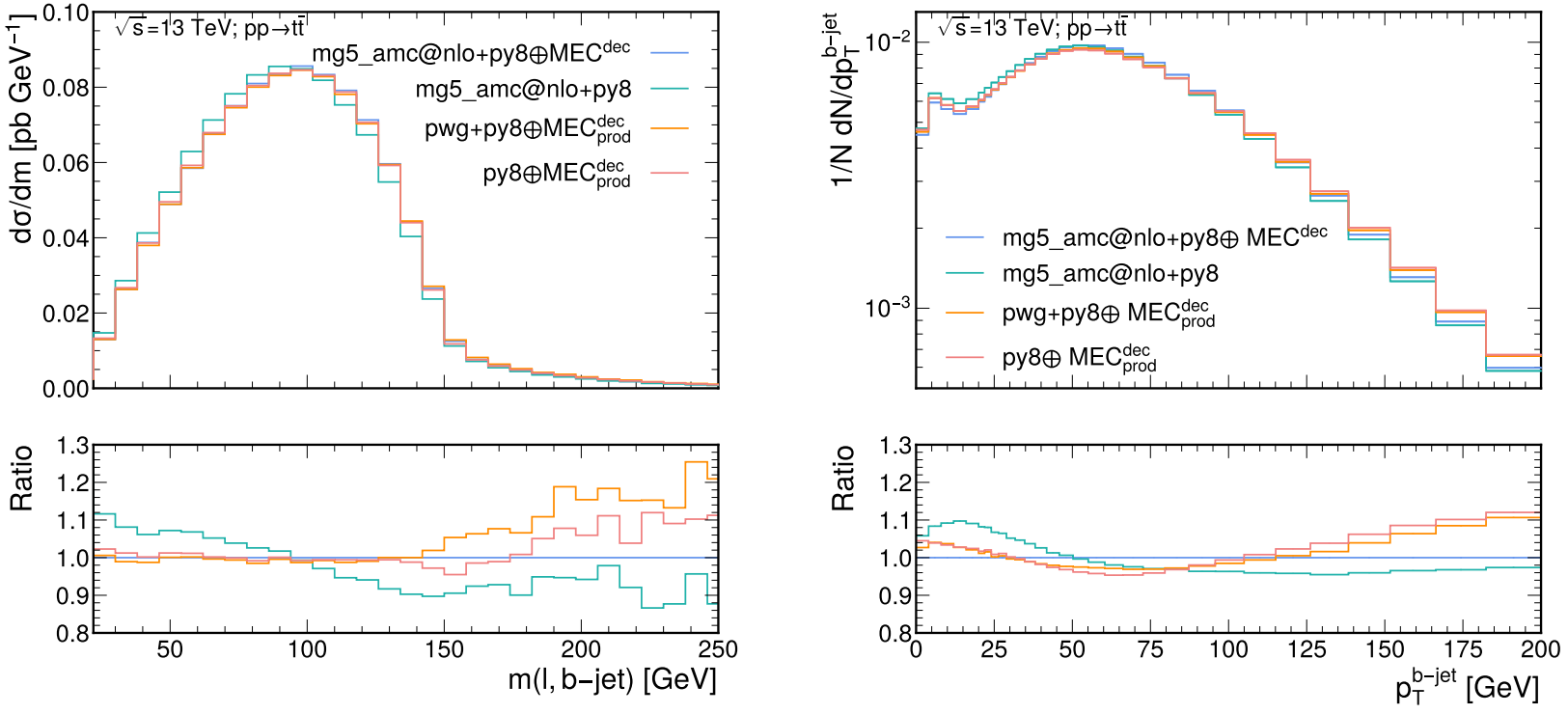


Figure: The distribution in the invariant mass of the (lepton,*b*-jet) pair (left panel) and the leading *b*-jet transverse momentum (right panel) in dileptonic $t\bar{t}$ events.

Difference? One Pythia setting

The Future: Why am I here?

ML & Pythia

ML $\equiv$ very complex parametrization of data

- Identify subtle patterns in complex data
- Automate tedious tasks and perform them rapidly

Lund string model is a probabilistic model

- Probabilities are about ignorance of causes
- Fundamental limitation (field theory + uncertainty principle) or lack of understanding / inability to calculate

Either way, ML can supplement string model to better fit data

Important: Hadronization uncertainties are large for many key measurements at HL-LHC

First Steps

Treat Lund String model predictions as DATA

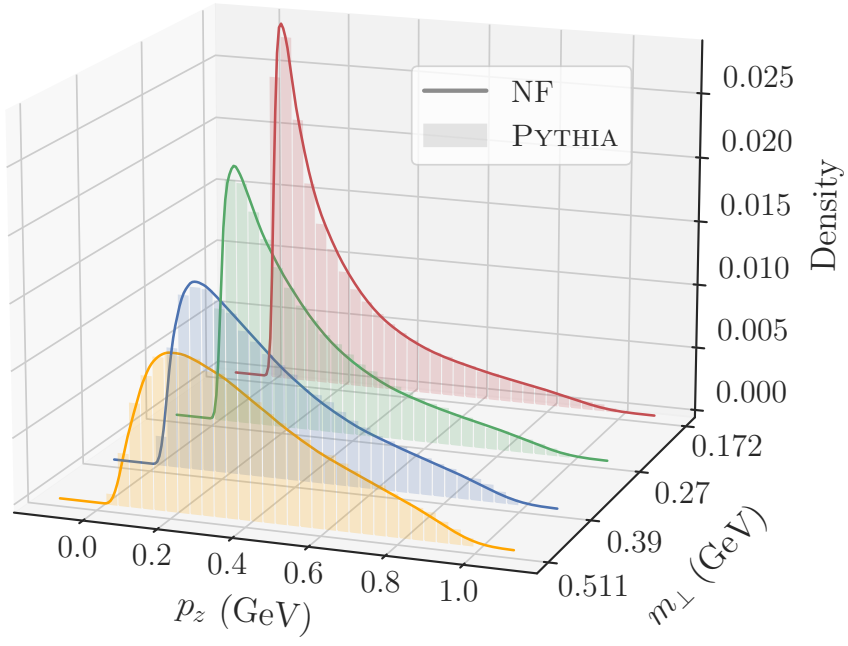


Figure: A comparison between the `Pythia` (histrograms) and `MLHAD` NF generated (solid lines) single emission p_z distributions conditioned over four values of transverse mass $m_\perp$ (left).

Next: Learn on data

- Develop hadronization-dependent observables (IR unsafe)

Concluding thoughts

- SM: highly mathematized description
- Not pure (requires phenomenological models) but *still* the most precise (and correct) model of natural phenomenon
- The Lagrangian is used to calculate. Nature does not calculate. Calculations are always incomplete. Certainly, Nature provides values for quantities, not bounds.

  The miracle of the appropriateness of the language of mathematics for the formulation of the laws of physics is a wonderful gift which we neither understand nor deserve.

  —*Eugene Wigner, The Unreasonable Effectiveness of Mathematics in the Natural Sciences (1960)*